

Investor Attention and Cryptocurrency Volatility: A Machine Learning and Econometric Analysis

Farid Gadirli^{a,*}, Minjia Chen^a and Getinet Haile^a

^a Nottingham University Business School, University of Nottingham, Jubilee Campus, Wollaton Road, Nottingham, NG8 1BB, UK

* Corresponding author

E-mail addresses: farid.gadirli@nottingham.ac.uk (F.Gadirli), minjia.chen@nottingham.ac.uk (M.Chen), getinet.haile@nottingham.ac.uk (G.Haile).

Investor Attention and Cryptocurrency Volatility: A Machine Learning and Econometric Analysis

Abstract

By distinguishing between individual and institutional investors, this study examines the role of investor attention in the realised volatility of the three main cryptocurrencies: BTC, ETH, and XRP. We utilise both machine learning and econometric techniques to comprehensively establish the link between investor attention and cryptocurrency volatility. Our results indicate that investor attention significantly predicts cryptocurrency volatility, with institutional investors demonstrating a particularly strong influence due to their access to substantial capital, advanced information, and relatively superior market analysis capabilities – including monitoring trading actions and analysing blockchain data. In turn, this enables them to potentially act as ‘whales’ in the cryptocurrency market. A comparison of empirical approaches reveals that machine learning techniques outperform GARCH-family models in predictive accuracy. By illuminating the role of investor heterogeneity and linking it to volatility, we address a crucial gap and contribute to the burgeoning cryptocurrency literature.

JEL: C10, C45, C53, G40, L12, L13

Keywords: Cryptocurrency; Investor attention; Realised volatility; Forecasting; Machine learning; GARCH-family models; Whales.

1. Introduction

The rapid and volatile growth of the cryptocurrency market in recent years has shown remarkable developments in this innovative digital platform (Makarov and Schoar, 2022). Since their inception over a decade ago, cryptocurrencies have consistently attracted intense speculative activity (e.g. Cheah and Fry, 2015), resulting in substantially higher price volatility *vis-a-vis* traditional financial assets (e.g. Biais et al., 2023). This enduring phenomenon has prompted strong concerns regarding excessive risks and bubbles associated with cryptocurrencies (Fry, 2018), presenting significant challenges for national regulators. Furthermore, cryptocurrencies, particularly those relying on proof-of-work protocol, have sparked enormous controversy due to their substantial resource consumption and detrimental environmental impacts. On the other hand, cryptocurrency, which is founded on fast-evolving technologies, including blockchains that are yet to be fully comprehended, provides lucrative opportunities to stakeholders. Investors, regulators and the broader public are currently engaged in vigorous debates on the integration of crypto assets into financial markets (Makarov and Schoar, 2022; White House, 2022; ESMA, 2024). Therefore, understanding the unique risk behaviour of cryptocurrencies has become essential for shaping effective investment strategies and regulatory frameworks. Despite extensive research on cryptocurrency volatility, the distinct roles played by different investor groups in influencing these dynamics remain insufficiently explored. This paper aims to address this gap by investigating the influences of institutional and individual investors on the volatility of cryptocurrencies, offering insights that contribute to both investment decision-making and regulatory policy development.

Extensive literature has focused on developing theoretical models concerning the risks and returns of cryptocurrencies and identified several groups of determining factors. First, the positive externality of network effects (e.g., Cong et al., 2021a; Biais et al., 2023) and the implications of production costs (e.g., Cong et al., 2021b; Sockin and Xiong, 2023) have been the subject of discussions in the literature. Second, exposure to other traditional assets (e.g. Schilling and Uhlig, 2019; Jermann, 2021) has been shown to be important. Third, empirical irregularities of cryptocurrencies, such as arbitrage opportunities (Makarov and Schoar, 2020) and manipulation (Gandal et al., 2018; Griffin and Shams, 2020), have been observed. Finally, an empirical analysis of a comprehensive set of pricing factors (Liu and Tsyvinski, 2021) has emphasised the significant roles of investor attention and exposure to network factors as predictors of future cryptocurrency returns while rejecting the influence of production factors.

Compared with traditional financial markets, the cryptocurrency market has consistently exhibited extraordinary volatility, driven by speculative trading and investor

sentiment alongside a lack of underlying business fundamentals (e.g. Cheah and Fry, 2015; Schilling and Uhlig, 2019)¹. Kogan et al. (2024) argue that the value of cryptocurrencies depends on investors' belief about the future market adoption of cryptos, which is largely signified by the presence of institutional investors. Dong et al. (2023) find evidence that institutional investors, who are always sensitive to monetary policy, transmit their volatile trading from the stock market to the crypto market. Individual investors are likely to interpret price movement as an indicator of changes in the probability of future adoption (Kogan et al., 2024). The volatilities of cryptocurrencies have also been found to be unusually asymmetric compared with equity markets (Baur and Dimpfl, 2018). Theoretical and empirical evidence has firmly established the influence of investor attention on investments in more traditional markets (Barberis et al., 1998), including stock market volatility (Andrei and Hasler, 2015), portfolio returns (Sicherman et al., 2016; Gargano and Rossi, 2018), and option returns (Choy and Wei, 2022). This relationship has also been found to vary across different national cultures (Cheon and Lee, 2017) and fluctuate over business cycles, significantly impacting asset prices, especially in the short term (Da et al., 2011).

The literature shows that volatility generally increases with investor attention (Andrei and Hasler, 2015). This relationship arises because investors, constrained by limited information, time, and cognitive abilities, exhibit bounded rationality in their decision-making, causing them to focus on specific attention-catching assets or those demonstrating salient features (Kahneman, 1973; Kahneman and Tversky, 1973; Grossman and Stiglitz, 1980; Conlisk, 1996, Barber and Odean, 2008; Bordalo et al., 2013). When they pay more attention to new market information, they tend to act immediately, causing buy pressure (Barber and Odean, 2008) and thereby inducing higher return volatility. Consequently, investors may require larger risk premiums to compensate for this attention-induced risk. Conversely, inattentive investors underreact to announcements (Dellavigna and Pollet, 2009), and therefore, the lack of attention might result in lower return volatility, thereby leading investors to require a smaller risk premium.

Not much is known concerning the link between investor behaviour and cryptocurrency volatility. One reason for the lack of empirical research in this area is the intrinsic pseudonymous nature of the cryptocurrency market. Traditional assets typically derive their value from fundamental factors, and the influence of investor attention is often endogenously

¹ Nevertheless, Biais et al. (2023) show that Bitcoin's fundamental value lies in its transaction costs and benefits.

connected with these underlying fundamentals. However, cryptocurrencies lack business fundamentals and thus provide an ideal setting to test the link with investor attention (Schilling and Uhlig, 2019; Pagnotta, 2022). Besides, Merkley et al. (2024) argue that crypto information is particularly in strong demand due to the large number of individual investors, which exceeded 110 million in 2022, and gives rise to information dissemination via social media, particularly X. Using Bitcoin, Shen et al. (2019) have shown that investor attention, as proxied by tweets, serves as a significant predictor of realised volatility. Expanding on this with a broader range of cryptocurrencies, Al Guindy (2021) and Smales (2022) confirm a positive relationship between investor attention and increased price volatility. Sockin and Xiong's (2023) theoretical pricing model shows that investor attention or speculator sentiment generates momentum in the cryptocurrency market, but the momentum effect diminishes when the model controls for attention and sentiment. Long et al. (2024) show that retail investors' sentiments on social media drive cryptocurrencies' intraday volatility, while Akyildirim et al. (2024) find news media sentiments play a significant role in cryptocurrency returns spillover. Further, investor attention is often measured in the literature by proxies such as internet searches (e.g., Liu and Tsyvinski, 2021), social media posts (e.g., Ozdamar et al., 2022), news stories (e.g., Barber and Odean, 2008; Andrei and Hasler, 2015), and trading account logins (e.g., Sicherman et al., 2016; Gargano and Rossi, 2018).

We contribute to the literature in four dimensions. Firstly, most research so far focuses primarily on Bitcoin, largely due to challenges related to data availability and the complexities of conducting in-depth analysis across multiple cryptocurrencies. We expand the coverage to also include ETH and XRP, which are both among the most established cryptocurrencies. Besides, while Bitcoin is primarily regarded as a store of value, the Ethereum network introduces the concept of a programmable blockchain that supports smart contracts and decentralised applications, and XRP is designed for fast, low-cost transactions. Even though XRP no longer consistently ranks among the top three cryptocurrencies by market capitalisation in the most recent years, its transactional purpose makes it fundamentally different from BTC and ETH.² Therefore, this selection allows for a comparative analysis of cryptocurrencies with distinct functionalities and use cases.

Secondly, we study the distinct effects of attention between individual and institutional investors. Since individual and institutional investors have primarily different motives and

² XRP has been a popular research object by scholars, e.g. Liu and Tsyvinski (2021) and Liu et al. (2022).

trading strategies (Nofsinger and Sias, 1999, Barber and Odean, 2000; Dennis and Strickland, 2002), it is plausible to expect they focus their attention and subsequent actions in distinctive ways. While the behaviour of both groups has drawn significant interest in academia, detailed comparisons between them remain surprisingly scarce. Nofsinger and Sias (1999) compare the relative importance of herding behaviour by institutional versus individual investors and find a strong positive correlation between changes in institutional ownership and stock returns. The authors conclude that institutional investors engage in greater positive feedback trading than individual investors, thereby their herding exerting a larger influence on returns. Another notable work in this realm is Barber and Odean (2008), who demonstrate that individual investors often exhibit attention-driven overtrading behaviour in the stock market, potentially inflating stock prices temporarily. However, this phenomenon is not observed among institutional investors, owing to their more sophisticated capacity for processing information and employing explicit purchase criteria. Focusing solely on individual Bitcoin investors, Hackethal et al. (2022) examine their various characteristics and investment behaviour and find that these investors are generally risk-loving, high-income males whose investments are often induced by high media sentiment. In their latest work, Akepanidaworn et al. (2023) identify that institutional investors demonstrate proficiency in buying and heuristics in selling, which they attribute to an asymmetric distribution of cognitive resources, i.e. attention. Furthermore, using three years of data across ten cryptocurrencies, Ozdamar et al. (2022) discovered that retail and institutional attention have differentiated effects on returns, idiosyncratic risk, and liquidity in the cryptocurrency market.

Białkowski (2020) argues that without effective safeguards such as stop-loss rules, cryptocurrencies characterised by extraordinarily high volatilities might not be favoured by institutional investors. Nonetheless, the cryptocurrency market has witnessed a remarkable surge in the presence of institutional investors within a relatively short span of time. For example, the latest report by PricewaterhouseCoopers (PwC) (2024) reveals that approximately half of the surveyed traditional hedge funds are now investing in digital assets, with the expectation of further growth in capitalisation.³ Moreover, collective holdings by financial institutions issuing Bitcoin exchange-traded funds (ETFs), such as BlackRock, Grayscale, Fidelity, and others, have recently surpassed a significant milestone of 1 million BTC valued

³“PwC 2024 Global Crypto Hedge Fund Report”, <https://www.pwc.com/gx/en/industries/financial-services/crypto-services/sixth-annual-global-crypto-hedge-fund-report.html>.

at over \$70 billion in May 2024.⁴ In addition, the approval of Ethereum ETF applications by the Securities and Exchange Commission in May 2024 set the stage for ETH ETFs to begin trading on the US stock market.⁵ These milestones not only signify the increasing acceptance of digital assets in mainstream finance but also provide investors with a new, regulated way to gain exposure to Bitcoin and Ethereum. On the other hand, due to the absence of national boundaries, the number of registered users on Binance, the world's largest cryptocurrency exchange, exceeded 250 million by the end of 2024 – potentially more than the number of traders in traditional financial exchanges.⁶ As inexpensive and accessible fintech tools continue to proliferate, the nexus of trading power and analytical capacity between individual versus institutional investors is fast evolving in the cryptocurrency market. This underscores the significance of our objective to differentiate between these two investor groups.

Thirdly, our study pioneers a novel investigation into the phenomenon of whales – dominant monopolistic investors with substantial holdings – in the cryptocurrency market. Unlike highly regulated traditional financial markets, the emergent and volatile nature of the cryptocurrency market has led to a high concentration of whale activity. Liu et al. (2023) highlight that in the Bitcoin market, a mere 1% of users contribute to over 95% of market volume, with over 90% of addresses having executed no more than two transactions in their lifetime and remaining largely inactive for years. Our extended dataset, spanning over a longer period than previous studies, reveals similar patterns for BTC and ETH. With their huge pseudonymous holdings and the lack of centralised accountability or disclosure requirements (Makarov and Schoar, 2022), whales possess the power to exert abrupt influence on the market through largely unrestricted trading. Therefore, understanding the monopolistic role of whales is vital for comprehending market risk and investor behaviour in the cryptocurrency sphere. To our knowledge, our research represents the first comprehensive exploration of the whale phenomenon across the top three cryptocurrencies and an examination of its link with investor attention and return volatility.

Finally, existing research has predominantly relied on conventional forecasting methods such as vector autoregression (VAR) models (e.g. Al Guindy, 2021). Our study

⁴ Yahoo finance news, “Bitcoin ETFs Now Hold 1 million BTC in Custody, Worth \$70.5 Billion”, <https://finance.yahoo.com/news/bitcoin-etfs-now-hold-1-051241521.html>.

⁵ Forbes Advisor, “What Does the ETH ETF Mean for Crypto”, <https://www.forbes.com/advisor/investing/cryptocurrency/what-eth-etf-approval-means-for-crypto/>.

⁶ Binance, “State of the Blockchain: 2024 Year in Review”, https://public.bnbstatic.com/reports/2024_EOY_Report.pdf.

advances the literature on investor attention and asset pricing by utilising cryptocurrency data to explore the link between investor attention and return volatility using both the Generalised Autoregressive Conditional Heteroskedasticity (GARCH) family models and machine learning techniques, including Nonlinear Autoregressive Neural Networks with Exogenous Inputs (NARX), Random Forests (RF), and Support Vector Regression (SVR). Unlike VAR models, GARCH methods explicitly model conditional variances of time series, accommodating time-varying volatility patterns. Machine learning methods can handle many complex and potentially endogenous predictor variables, often enhancing predictive performance in nonlinear and high-dimensional settings. In other words, GARCH and machine learning methods provide a more adaptable approach for modelling volatility and complex dependencies, making them well-suited for this analysis compared to traditional VAR models. Based on this framework, we hypothesise that increased investor attention leads to heightened cryptocurrency return volatility.

Our findings indicate that institutional investor attention significantly and consistently predicts cryptocurrency volatility, while individual investor attention exhibits variable links depending on the nature of the cryptocurrency considered. By distinguishing between individual and institutional investor attention, we emphasise the importance of identifying risk exposure in the cryptocurrency market. Furthermore, our study highlights the significant role of whale behaviour in influencing market dynamics. Our analyses are crucial for shaping effective regulatory regimes for both national and international financial authorities.

2. Empirical Methodology and Data

2.1 Empirical Methodology

In this study, we compute a 30-day rolling realised volatility (RV). It is defined as the standard deviation of the daily continuously compounded returns over a 30-day period.⁷

The econometric analysis implements GARCH family models with exogenous variables (i.e., GARCH-X) using Stata⁸. Several recent studies have highlighted the advantages and extensive application of GARCH family models in analysing cryptocurrency volatility

⁷ $RV = \sqrt{\sum_{t=i-N+1}^i (r_t - \bar{r}_i)^2 / (N - 1)}$, where $N = 30$, is the number of days in the rolling window, i is the current day, $\bar{r}_i$ is the average return over the past N days up to day i , r_t is the return at day t and equals to: $r_t = \ln(P_t) - \ln(P_{t-1})$ with P being price. RV is widely used in finance as a measure of historical market volatility and is adopted by major financial and cryptocurrency analytics platforms, including Coinmetrics.io (2024).

⁸ StataCorp. 2023. Stata Statistical Software: Release 18. College Station, TX: StataCorp LLC.

(Umar et al., 2021; Pattnaik et al., 2023; Zhang et al., 2023). Building on the potential for improved results through the inclusion of explanatory variables (Walther et al., 2019), our econometric framework involves GARCH family models with exogenous variables (i.e., GARCH-X).⁹ Recognising the intricate dynamics of cryptocurrencies, we evaluate several GARCH-X (p, q) models, ranging from a parsimonious (1,1) to a more complex (3,3), where p and q respectively denote the number of GARCH and ARCH lags. GARCH-X models have recently attracted significant attention in the literature, with their properties comprehensively discussed by Han and Kristensen (2014), Francq and Thieu (2019), and Pedersen and Rahbek (2019). We select the optimal GARCH-X models for each cryptocurrency using standard model selection criteria, including log-likelihood, Akaike Information Criterion (AIC), Schwarz-Bayesian Information Criterion (BIC), the significance of the GARCH/ARCH components, residual diagnostic tests, and prediction accuracy. Model accuracy is assessed by comparing deviations from realised volatility using root mean square error (RMSE) and mean absolute error (MAE).

We also utilise machine learning techniques, including Nonlinear Autoregressive Neural Networks with Exogenous Inputs (NARX), Random Forests (RF), and Support Vector Regression (SVR), implemented in MATLAB¹⁰. The NARX model is a specialised type of recurrent dynamic network that is commonly used for time series prediction. It extends the linear autoregressive model with exogenous inputs (ARX) by incorporating nonlinearity and feedback connections, enabling it to capture complex dynamics in data sequences and enhance predictive performance (MathWorks, 2024). The NARX model is expressed as (Lin et al., 1996; Ruiz et al., 2016; MathWorks, 2024):

$$y(t) = g(y(t-1), y(t-2), \dots, y(t-p), x(t-1), x(t-2), \dots, x(t-k)) + \varepsilon(t) \quad (1)$$

where $y(t)$ is the output of the system showing the predicted values obtained from a nonlinear network function $g(\cdot)$ of previous p values of the target series $y(t-1), y(t-2)$

⁹ We assess both the standard GARCH-X model and asymmetric variants, such as exponential GARCH (EGARCH), threshold GARCH (TGARCH), the GJR form of threshold GARCH (GJR-GARCH), and asymmetric power GARCH (APGARCH), all incorporating exogenous variables. To explore the potential for enhancing predictive performance, we also run models with and without ARIMA components. The lag order selection for the ARIMA component is guided by the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots, while also considering computational expenses.

¹⁰ The MathWorks Inc. 2024. MATLAB version: 24.2.0 (R2024b), Natick, Massachusetts: The MathWorks Inc. <https://www.mathworks.com>.

,..., $y(t - p)$, known as feedback delays, and previous k values of an exogenous input series x ($t - 1$), $x(t - 2)$,..., $x(t - k)$, referred to as input delays, with p and k representing the number of time delays for the feedback and input, respectively, and $\varepsilon(t)$ is an error term. The function $g(\cdot)$ is approximated during training through MATLAB's network weight and neuron bias optimisation process (Ruiz et al., 2016).

The neural network architecture used in this study consists of an input layer, a hidden layer and an output layer.¹¹ The input layer consists of source nodes, where each node represents a predictor variable in the model. The hidden layer applies nonlinear transformations to the input data using weights and biases, which are optimised during the learning process to enable the network to model complex patterns in the data (Haykin, 2009). Following Gu et al. (2020) and the recommendations of MathWorks, inputs and targets are normalised to the range $[-1,1]$ prior to training (MathWorks, 2024).

To ensure a sufficiently large training set and proper model assessment, MATLAB's default partition scheme is adopted, splitting the data into 70% for training, 15% for validation, and 15% for testing (MathWorks, 2024). Specifically, the training subset is used for learning and fitting model parameters, the validation subset is used to evaluate performance and guide hyperparameter adjustments, and the testing subset — completely isolated from both training and validation — provides a final measure of the model's true predictive power (James et al., 2017; Gu et al., 2020).

To address the black-box nature of neural networks, multiple networks or topologies are constructed and compared to identify the best models. Adhering to the principle of parsimony, we use a bottom-up approach, gradually increasing the width rather than depth of the network to avoid issues like the vanishing gradient problem. Since our objective is to model volatility rather than focus on multi-step ahead predictions, the networks are run in an open-loop configuration. This approach uses true outputs instead of feeding back estimated outputs, enhancing input accuracy and simplifying the architecture into a pure feedforward design, which enables more efficient training (Menezes and Barreto, 2008; MathWorks, 2024). Under this scenario, the neural network architecture is similar to those used in Gu et al. (2020), Bianchi et al. (2021), Leippold et al. (2022) and Avramov et al. (2023). We select the optimal

¹¹ An input layer includes previously selected independent variables (predictors), while the output layer represents the target variable or realised volatility.

model based on the lowest validation error while also considering error deviations¹² across training, validation, and test subsets to mitigate overfitting and underfitting issues.

The support vector regression algorithm employed in this study is derived from support vector machines. SVR is particularly effective for financial time series predictions due to its ability to model nonlinear processes via kernel mapping without requiring prior knowledge about the data (e.g., Khashanah and Shao, 2022).

Given training data (x_t, y_t) where $x_t \in \mathbb{R}^p$ an input vector and $y_t \in \mathbb{R}^1$ an output scalar, the basic SVR function is expressed as (Awad and Khanna, 2015):

$$f(x) = w^T \varphi(x) + b \quad (2)$$

where $f(x)$ is the sample regression function, w is the weight vector with w^T denoting its transpose, x represents an input vector, and b a bias term. The nonlinear mapping function $\varphi(x)$ transforms the original input data vector into a higher-dimensional feature space, enabling the data to become linearly separable (Schölkopf and Smola, 2001; Stasinakis et al., 2016; Khashanah and Shao, 2022). The estimated parameters w and b define the hyperplane for the regression function $f(x, w)$. SVR employs an ε -insensitive loss function, which penalises predictions only if they deviate by more than ε from the true output. The loss function is defined as (Vapnik et al. 1997; Murphy, 2012; Awad and Khanna, 2015):

$$L_\varepsilon(y, f(x, w)) = \begin{cases} 0 & \text{if } |y - f(x, w)| \leq \varepsilon \\ |y - f(x, w)| - \varepsilon & \text{otherwise} \end{cases} \quad (3)$$

where ε is a non-negative value defining a tube around the true value, within which predictions are considered acceptable and not penalised. Data points lying on or outside the ε margin act as support vectors and form part of the function $f(x, w)$ (Chen et al., 2010). Estimators of w and b are obtained by solving the primal constrained optimisation problem:

$$\min \frac{1}{2} \|w\|^2 + C \sum_{t=1}^n (\xi_t + \xi_t^*)$$

Subject to:

$$y_t - w^T \varphi(x_t) - b \leq \varepsilon + \xi_t \quad t=1, \dots, T$$

¹² The deviation boundaries are determined in relation to the general error levels of the models.

$$\begin{aligned}
w^T \varphi(x_t) + b - y_t &\leq \varepsilon + \xi_t \quad t=1, \dots, T \\
\xi_t, \xi_t^* &\geq 0 \quad t = 1, \dots, T \\
C &> 0
\end{aligned} \tag{4}$$

where C is a tuneable regularisation parameter which controls the trade-off between minimising error and model complexity. The slack variables ξ_t and ξ_t^* measure deviations from the upper and lower bounds of the ε -margin, respectively. To solve this quadratic optimisation problem efficiently, the Lagrange dual formulation is applied, involving two Lagrange multipliers a_t, a_t^* and inner product kernel function $K(x_t, x)$ to facilitate data mapping (Schölkopf and Smola, 2001; Chen et al., 2010). By incorporating the Karush–Kuhn–Tucker (KKT) complementarity conditions, the decision function in equation (2) yields (Haykin, 1999; Stasinakis et al., 2016; Khashanah and Shao, 2022):

$$f(x) = \sum_1^n (a_t - a_t^*) K(x_t, x) + b \tag{5}$$

where $a_t, a_t^* \in [0, C]$ and $K(x_t, x) = \varphi^T(x_t) \varphi(x)$. We select an optimal kernel function by evaluating all available options in MATLAB's Regression Learner App and follow methods by MathWorks (2024).¹³ To avoid overfitting and evaluate the performance of predefined hyperparameters, k -fold and holdout cross-validation techniques are applied. In k -fold validation, the data is divided into k randomly selected folds (or subsets) of roughly equal size, training on $k-1$ subsets while using the remaining subset for validation (Duan et al., 2003). The process is repeated k times, and the mean validation error is reported as the evaluation metric. Considering computational costs, a 5-fold cross-validation approach is chosen to achieve a balance of training speed and reliable performance estimates, which is also MATLAB's default option. For holdout validation, in order to have a similar partition as in the neural networks, 70% of data is allocated for training, 15% for validation and 15% for testing.

The random forests algorithm estimates an outcome value by averaging predictions from an ensemble of randomly constructed decision trees. Each tree is trained on a bootstrap sample of the data and built using recursive partitioning with randomly selected features.

¹³ MATLAB has several built-in kernel functions, including linear, polynomial, and Gaussian or Radial Basis Function (RBF). By default, the standardisation of variables is set to true, and they are centred and scaled by the corresponding weighted column mean and standard deviation (MathWorks, 2024). The best kernel functions and hyperparameters are selected based on validation RMSE.

Considering n independently and identically distributed random samples indexed by $i = 1, \dots, n$, predictions for each data point x are obtained by averaging outputs from an ensemble of decision trees (Breiman, 2001; Wagner and Athey, 2018; Athey et al., 2019). In its simplest form, the RF regression can be viewed as achieving a forest-based prediction, $\hat{y}(x)$, which is obtained based on averaging N tree-based predictions $\hat{y}_n(x)$ from the ensemble of trees as:

$$\hat{y}(x) = N^{-1} \sum_{n=1}^N \hat{y}_n(x) \quad (6)$$

As Athey et al. (2019) observe, for conditional mean estimation, this averaging represented by equation (6) is equivalent to the weighting-based estimation used in forests. We use the same k-fold and holdout validation approaches to evaluate the RF model. To identify an optimal complexity level that avoids overfitting and excessive computation time, we experiment with two parameters: the minimum leaf size, which specifies the minimum number of observations required to form a leaf node, and the number of learning cycles, which determines how many decision trees the ensemble will contain.

In addition to comparing models with and without the additional control variables, we also implement the SHapley Additive exPlanations (SHAP) technique to interpret the significance of investor attention in the prediction of the SVR and RF models.¹⁴ SHAP has gained prominence more recently in interpreting machine learning models (Lundberg and Lee, 2017; Wang et al., 2023; Bai and Zhang, 2024). SHAP values compute the average marginal effects of each feature across all possible permutations, thereby quantifying the contribution of each feature on the model's forecast (Aboze, 2023; Gnecco et al., 2023; Molnar, 2023). We apply the Kernel SHAP algorithm to the training data while optimising computational efficiency by incorporating parallel processing and capping the maximum subset size at 2^5 for models with additional predictors, following the suggestions of MathWorks¹⁵ (Aas et al., 2021; Gnecco et al., 2023; MathWorks, 2024).

The significance of predictors is evaluated using two complementary approaches. First, Shapley importance plots are based on the average absolute SHAP values across all observations, which provides a ranking of features based on their global importance to the

¹⁴ MATLAB is yet to develop SHAP functionality for NARX models.

¹⁵ Computing exact Shapley values requires evaluating 2^M subsets, where M is the total number of features, leading to exponential computational complexity. To mitigate this, a cap of 2^5 is applied, limiting the number of evaluated subsets. Exponent 5 represents the maximum number of features included in any evaluated subset.

model. This approach allows us to estimate each feature's contribution by analysing its marginal impact on model predictions across all possible feature combinations. Second, we employ summary plots to visualise both a feature's importance and its directional effects on model prediction, depicting the distribution of SHAP values for each feature (Aboze, 2023; Molnar, 2023; MathWorks, 2024).

To evaluate model performance, we compare the econometric and machine learning models using RMSE and MAE measures. These measures provide a robust basis for assessing predictive accuracy and allow for a systematic comparison of methodologies.

2.2 Data

We use daily price data for BTC, ETH, and XRP, with an overall sample period from 1st January 2014 to 5th May 2023, sourced from Coinmetrics.io. Collectively, they constitute 73% of cryptocurrency market capitalisation as of 5th May 2023. The dependent variable in all cases is realised volatility. Figure 1 plots the realised volatility of each cryptocurrency over the sample period, where XRP exhibits the highest level of volatility, followed by ETH and BTC. Figure 2 provides insights into the trend of cryptocurrency holdings by the top 100 addresses, addresses holding at least 0.1% of supply, and those holding at least 0.01% of supply. Generally, BTC and XRP show a declining supply share of the top 100 addresses, whilst ETH exhibits an upward trend, surpassing its 2018 level by 2023.

To capture investor attention, we use data from Bloomberg Terminal and X (formerly Twitter) platform. Bloomberg Terminal news is shown to be an effective indicator in capturing institutional investors' attention, including in the crypto market (Ben-Rephael et al., 2017; Blankespoor et al., 2020; Chiu et al., 2021; Iliev et al., 2021; Ozdamar et al., 2022). Accordingly, we measure institutional attention using the Bloomberg Terminal news trend function, which tracks the daily count of cryptocurrency-related stories. For individual attention, we rely on a daily number of tweets on X for each cryptocurrency, as it serves as a key platform where cryptocurrency investors engage and influence others, potentially reflecting the intensity of retail investor interest (Shen et al., 2019; Al Guindy, 2021; Ozdamar et al., 2022; Merkley et al., 2024). Figure 3 shows the trends in individual and institutional investor attention. Initially, cryptocurrencies attracted low levels of attention but have significantly increased over time, with a sharp surge during the Bitcoin price spike to nearly 20,000 USD in December 2017. As well as attention measures, our empirical analyses control

for cryptocurrency-specific, network, financial market, and uncertainty factors¹⁶. While we use several metrics for the descriptive analysis of wealth concentration (whale effect), the empirical analysis relies on two proxies: the number of addresses with a balance greater than or equal to 0.1% and 0.01% of the total supply. These thresholds are chosen because unit-based measures require different cutoffs for each cryptocurrency due to price and supply variations, whereas these metrics provide a standardised approach across the analysed cryptocurrencies. This method enhances comparability, though we acknowledge that differences in supply distribution mechanisms across cryptocurrencies may still influence the effectiveness of these proxies.

The descriptive statistics presented in Table 1 summarise the distinctive characteristics of the three cryptocurrencies examined. In terms of average closing prices over the sample period, ETH's value is approximately 10% of BTC, while XRP's value is a mere 0.004% of BTC. A similar pattern is observed for average returns, with BTC yielding the highest returns, followed by ETH, while XRP reports negative mean returns. On the other hand, BTC displays the lowest level of volatility, followed by ETH and XRP, indicating varying risk profiles across them.

Individual investors exhibit the highest attention to ETH, followed by Bitcoin and XRP, whereas institutional investors predominantly focus on BTC, with less attention to ETH and XRP. XRP consistently attracts the least attention from both groups. A comparison of the two investor categories revealed that, on average, individual investors exhibit a greater daily attention level to these cryptocurrencies. These observations suggest distinct behavioural patterns between the two groups of investors regarding cryptocurrency investments. The variation in attention among cryptocurrencies is relatively lower for individual investors, whereas institutional investors show a more pronounced preference, with Bitcoin consistently dominating over the years. However, there has been a notable increase in institutional attention towards ETH in more recent years.

Our data reveal a high concentration of cryptocurrency ownership, with the top 1% of addresses controlling between 87.6% and 97.3 % of the market, as detailed in Table 2. This indicates the dominance of certain groups, individuals or institutions within the cryptocurrency markets. On the other hand, XRP exhibits less concentration with broader participation,

¹⁶ For the full list of variables, their definitions, and sources, please refer to the appendix. Our initial set of control variables included additional factors such as exchange rates, commodity prices, the FED funds rate, trading volume, and momentum. However, the number of controls was reduced due to challenges in maximising likelihood estimates in econometric models with complex structures, convergence issues, and failures in adequacy tests. Consequently, the final analysis includes only variables that contribute meaningfully to predictive performance, ensuring no significant loss in explanatory power.

implying that only substantial holdings can exert considerable market influence. In contrast, BTC and ETH markets exhibit higher concentration levels, meaning that a small group of individuals or institutions control large portions of cryptocurrencies.

These findings underscore notable differences in pricing, volatility, market structure, and investor behaviour across the cryptocurrency market.

3. Results and Discussion

3.1 GARCH-X

As seen in the accuracy measures section of Table 3, XRP is the least predictable, exhibiting the highest errors, followed by ETH and BTC. This aligns with the volatility rankings of these cryptocurrencies. Bitcoin's relatively lower volatility compared to other cryptocurrencies could be attributed to its established status, wide adoption, large market capitalisation, high liquidity, and increasing institutional involvement. In contrast, altcoins like ETH and XRP could exhibit higher volatility due to speculative interest and sensitivity to external factors such as regulatory changes, protocol updates, and specific developments in their blockchain networks. These altcoins are often used for various purposes¹⁷ and thus have different investor bases, contributing to their volatility.

Our analyses indicate that both institutional and individual investors influence cryptocurrency volatility. Tables 3-5 reveal that institutional investor attention has a statistically significant impact on volatility across all GARCH-X specifications. This robust finding is intuitive, and we thus argue that institutional investors, through large transactions, play a crucial role in driving market price movements. Their market entry and exit are likely to be seen as important signals by other market participants, thereby leading to rapid price movements as others react to the actions of these significant players. Additionally, these

¹⁷ More details can be found on the official websites of ETH <https://ethereum.org/en/eth/> and XRP <https://ripple.com/xrp/>.

findings highlight the possibility of institutional investors destabilising the market by triggering herding (Scharfstein and Stein, 1990; Cai et al., 2019), speculation (Brunnermeier and Nagel, 2004; Jang and Kang, 2019) and noise trading (Trueman, 1988; Shleifer and Vishny, 1997; Ben-David et al., 2021).

Herding and noise trading might also be attributed to unequal wealth distribution and the presence of whales, including high-net-worth individuals, early adopters, enthusiasts, exchanges and institutional investors. Our whale effect analysis (Tables 4 and 5) finds whales with holdings of at least 0.01% of supply exert significant impacts for BTC, while for ETH, the significant threshold is at least 0.1% of supply. Additionally, none of the whale effect measures are found to have a significant link with XRP. These findings confirm higher market concentration in the BTC and ETH markets. Due to price differences, a lower threshold (0.01% of supply) is already capable of impacting BTC volatility, whereas more holdings (0.1% of supply) are needed to affect the ETH market.

The insignificance of a higher threshold in BTC also signals that major BTC holders, such as early adopters and enthusiasts, often adopt long-term holding strategies and focus on market development and stability rather than active trading, despite their potential to influence the market. On the other hand, the insignificant whale effects for XRP could be attributed to its ownership structure. A substantial portion of XRP is owned by Ripple Labs, its co-founders, and is held in escrow accounts, all of which are not actively used in trading.¹⁸ This implies low market concentration and a broader distribution of holdings among the rest of the XRP holders.

It can be observed that the statistically significant whale and institutional investor variables both have a positive sign, indicating that each of them contributes to increasing cryptocurrency volatility. Notably, the higher magnitude of the whale activity coefficient underscores the greater potential of whales to destabilise the market compared to other market participants. It is important to recognise that whales can include not only individual investors but also institutions that hold substantial cryptocurrency positions. Therefore, it can be suspected that institutions might act as whales or mimic them, influencing the market through large-volume transactions. However, the lack of a significant interaction between institutional attention and whale activity suggests their effects on volatility operate independently. While institutions may occasionally mimic whale behaviours, they could operate through distinct

¹⁸ “Q4 2024 XRP Markets Report”, <https://ripple.com/insights/q4-2024-xrp-markets-report/>.

informational mechanisms guided by market analysis, sentiment tracking and risk management principles rather than purely following the transactions of other whales.

On the other hand, the insignificance of this interaction may also reflect the diverse characteristics and behaviours among whales, who have the potential to engage in diverse investment strategies, ranging from long-term holding — driven by a belief in the future of blockchain technology and digital assets — to short-term gains and speculative trading, which may further explain their independent influence from other institutions. Moreover, even when institutions are acting as whales, their influence on volatility can operate independently of the general level of institutional attention in the market.

The destabilising behaviour of whales or institutions in the cryptocurrency market resembles imperfect market competition, where certain groups have some degree of market power while the barriers to entry and exit are not so high. The assumption of imperfect market competition helps to explain institutions' seemingly “irrational” strategies. In the context of imperfect competition among informed traders, those that use heuristics and tend to overreact to their private information can achieve higher expected utility than their Bayesian counterparts, as their overreaction leads them to trade more aggressively (Palomino, 1996; Kyle and Wang, 1997; Fischer and Verrecchia, 1999). While Bayesian investors may dominate in a perfectly competitive market by making statistically optimal portfolio rebalancing decisions that drive heuristic behaviours out, in markets with imperfect competition, where large investors can influence prices, heuristic strategies may gain a competitive edge or coexist with Bayesian strategies, particularly when these investors possess superior private information (Verrecchia, 2001). Nevertheless, their survival depends on the market not being Bayesian on average, a scenario that might also be observed in the cryptocurrency markets.

An example of an information advantage that can lead institutions to behave in ways that may appear “irrational” is their superior access to market data and relatively advanced analytical skills (Brunnermeier and Nagel, 2004), potentially allowing them to analyse cryptocurrency market sentiment, examine blockchain networks, and monitor the activity of whales or other high-profile investors – capabilities that can be quite costly and computationally demanding for individual investors. Following such information, institutions may mimic whales or even herd and act as whales to get higher utility, even if such decisions seem to be irrational, leading them to engage in noise trading and speculative bubbles, which could contribute to market instability and deviations from fundamentals (De Long et al. 1990; Abreu and Brunnermeier, 2002; Griffin et al., 2003; Sias and Whidbee, 2010).

For instance, conventional theoretical literature explains institutional herding by informational cascades, where institutions deduce information from each other's trading behaviour, particularly their better-informed peers (Banerjee, 1992; Bikhchandani et al., 1992; Sias, 2004), investigative herding when they are following the same indicators and as a result receiving positively correlated information (Froot et al., 1992; Hirshleifer et al., 1994) and characteristic herding when institutions are attracted to the assets with certain characteristics such as liquidity and volatility (Falkenstein, 1996; Del Guercio, 1996; Gompers and Metrick, 2001; Bennett et al., 2003). Another reason for the potential herding behaviour of institutions and their desire to mimic other actors can be the significant reputational costs they face if their investment strategies diverge from those of their peers (Scharfstein and Stein, 1990; Trueman, 1994; Sias and Whidbee, 2010). Consequently, if institutional investors perceive noise trading and speculative behaviour as rational, they may exploit these as market imperfections for potential gains.

On the other hand, the impact of individual investor attention on volatility varies by the type of cryptocurrency and the level of cryptocurrency holding. Specifically, individual investor attention is statistically significant for BTC and XRP when control variables are used with holdings of 0.1% of supply or more (Table 4) and 0.01% of supply or more (Table 5). This contrasts with the insignificant coefficients in Table 3 without controls. No significant link is found between individual investor attention and ETH volatility, suggesting that despite a high number of individual investor attention, this cryptocurrency market is dominated by institutional investors, who potentially use the Ethereum network not only for cryptocurrency transactions but also as a platform for executing smart contracts and hosting decentralised applications. The marginally significant positive association between individual investor attention and the volatility of BTC suggests that the trading behaviour of individual investors is largely uninformed and represents noisy trading, which can be linked to the concepts of limited attention (Kahneman, 1973), information asymmetry (Hvidkjaer, 2008), salience (Bordalo et al., 2013; Cosemans and Frehen, 2021) and overconfidence (Daniel et al., 1998; Peng and Xiong, 2006). Conversely, the negative association between individual investor attention and XRP volatility may indicate that increased interest and participation from retail investors contribute to a more stable trading environment. Compared to institutional investors, whose large trades significantly impact market prices, retail investors' smaller transactions may have limited influence on volatility. Moreover, due to XRP's low price, individual investors are often motivated by long-term holding strategies based on assets' future potential rather than short-term trading gains.

Finally, the GARCH-X results in Tables 4 and 5 highlight additional factors that are important in predicting cryptocurrency volatility. Cryptocurrency network factors and financial market factors, particularly VIX, are shown to have a statistically significant link with the volatility of cryptocurrencies considered.

3.2 Machine Learning

Performance statistics from the NARX models are reported in Tables 6 and 7. Table 6 focuses solely on investor attention measures, while Table 7 controls for additional factors, including measures of whale effects.

Accuracy statistics, including the mean absolute error and root mean square error, are reported for training, validation and test samples. Comparisons of the statistics from the training sample *vis-à-vis* those from the validation and the test performance samples suggest no significant differences among the three samples in terms of error size. This suggests that we can be reasonably confident in ruling out overfitting, a common concern in machine learning analysis.

The significantly lower accuracy statistics, viz., mean absolute error and root mean square error, from NARX in Tables 6 and 7 are superior *vis-à-vis* the corresponding GARCH models in Tables 3 to 5, respectively. In other words, NARX demonstrates superior suitability for forecasting cryptocurrency volatility than the standard GARCH models, which have long been the workhorse of volatility modelling in traditional econometrics.

Importantly, the accuracy statistics in Tables 6 and 7 are remarkably similar. Given that the statistics in Table 6 are based on investor attention measures only while those in Table 7 include a range of additional controls, including the number of addresses with the largest holding of cryptocurrency, the similarity in these statistics suggests that investor attention is a significant predictor of cryptocurrency volatility. This is particularly evident for BTC, where the similarity in accuracy statistics highlights the strong predictive power of attention measures.

Alternative machine learning models, including support vector regression and random forests in appendix Tables A3-A6, generally corroborate findings from the NARX models. Among the models, NARX performs the best, generally followed by RF, SVR, and GARCH. This is in line with the research of Peng et al. (2018) and D'Amato et al. (2022), who highlight the superiority of machine learning based methods compared to traditional econometric methods in volatility forecasting. Across the alternative methods we have used, our results indicate that institutional and individual investor attention have been shown to be significant predictors of cryptocurrency volatility. The link found is more prominent for BTC and ETH,

highlighting the greater impact of investor attention on these cryptocurrencies. On the other hand, XRP exhibits a distinct pattern, where additional predictors contribute more to accuracy improvement, indicating a weaker influence by investor attention.

As noted earlier, we implement the SHAP technique to further interpret the significance of investor attention. Figures 4 and 5 present the SHAP plots for RF and SVR models using only institutional and individual attention measures for the three cryptocurrencies. The bar plots rank features by the mean of their absolute SHAP values, with higher values indicating greater importance. Accordingly, institutional attention has the most significant impact on the volatility of BTC and ETH, while individual attention is more influential for XRP. The summary plots combine feature importance with their corresponding effects, showing the distribution of SHAP values for each feature. Feature values are represented by a gradient shading scheme, with lower values in black and higher values in shades of grey and white. The plots reveal a wide spread of features, indicating that their influence varies considerably across different instances. Notably, the dominance of the grey and white gradient highlights the importance of increased institutional attention in influencing the volatility of the analysed cryptocurrencies. In contrast, while an increase in individual investor attention has been found to have a greater impact on BTC volatility, the effect appears to be the opposite for ETH and XRP. In other words, a decrease in individual investor attention appears to exert a greater influence on volatility than an increase. These findings highlight the complex interplay between different types of investors and their market behaviour.

SHAP plots for RF and SVR models with additional controls are provided in Appendix Figures A1 and A2, respectively. They exhibit remarkably similar patterns across models regarding the importance of our attention measures, with institutional attention playing a greater role in the volatility of BTC and ETH, while XRP's volatility is less influenced by these measures. Institutional investor attention is found to be even less important in its influence on XRP volatility. For XRP, other predictors — such as market capitalisation, active address count, and mean transaction fees — consistently rank higher in importance across all models, with the volatility index (VIX) and transaction count also surpassing attention measures in certain models.

XRP's unique ecosystem characteristics likely explain its reduced sensitivity to investor attention. The valuation of BTC is heavily influenced by growing demand, its legalisation processes, and speculative narratives around its role as a digital store of value. Similarly, ETH benefits from its integration into decentralised finance and smart contract ecosystems, which drive transaction volume and enhance network utility. In contrast, XRP's primary function lies

in facilitating cross-border payments for financial institutions. This focus on practical utility creates a more stable demand base, as financial institutions prioritise operational efficiency over speculation. As a result, institutional investments in XRP are often motivated by its role in liquidity optimisation and cost-effective remittance solutions. These factors contribute to a steadier demand profile for XRP, making it less reactive to fluctuations in investor sentiment.

Moreover, Ripple's dominant role in XRP's ecosystem, coupled with legal and regulatory scrutiny, dampens speculative interests. Ripple's significant control over the entire supply of XRP (100 billion tokens), through pre-mining and periodic escrow releases, generates scepticism among crypto enthusiasts who often prioritise decentralisation. This scepticism, in turn, limits volatility typically driven by scarcity narratives. Ripple's controlled supply strategy promotes market stability, further reducing the influence of investor-driven hype.

Overall, our empirical results affirm the critical role of investor attention, particularly from institutional investors, as an important predictor of volatility for BTC and ETH. For XRP, structural differences reduce the influence of attention-driven factors, emphasising the multifaceted interactions within the market and the unique characteristics of these cryptocurrencies.

4. Conclusion

The study explores the effect of investor attention on cryptocurrency volatility empirically. Our data spans three prominent cryptocurrencies, BTC, ETH, and XRP, over the period 1st January 2014 to 5th May 2023. The paper employed both traditional econometric models (viz., GARCH-X) and alternative machine learning techniques in the form of NARX, RF, and SVR. In addition, we apply the SHAP technique to interpret the significance of investor attention in cryptocurrency volatility.

Our findings reveal several critical insights. First, investor attention measures serve as significant predictors of cryptocurrency volatility, with institutional investor attention exerting a particularly strong influence. This is thought to be due to institutional investors' access to substantial capital, sophisticated information and relatively superior market analysis capabilities, including sentiment tracking, trade monitoring, and blockchain data analysis, which are likely to position some institutions as major market participants (whales). Second, institutional investor attention predominantly increases volatility across the three cryptocurrencies studied, whereas individual investor attention demonstrates mixed effects. Third, we reveal significant unequal wealth distribution in the cryptocurrency market, where

certain groups, individuals, or institutions dominate. Fourth, we explain the destabilising behaviour of whales or institutions through the lens of imperfect market competition, which enables them to exploit information asymmetries and amplify market inefficiencies. In doing so, whales and institutions not only create an environment that fosters herding behaviours – such as informational cascades, investigative herding, characteristic herding, and reputational herding – but also actively participate in these same herding dynamics, as established in the conventional theoretical literature. These results underline the intricate role of investor types in shaping market behaviours. Furthermore, the comparison of empirical approaches revealed that machine learning techniques outperform GARCH models in their predictive accuracy by better capturing the complex, non-linear relationships inherent to cryptocurrency markets. By illuminating the role of investor heterogeneity and linking it to volatility, this study has managed to address a crucial gap in the burgeoning cryptocurrency literature.

Our findings provide crucial insights into the evolving cryptocurrency landscape. They highlight the need for tailored approaches when analysing volatility across different cryptocurrencies, recognising the importance of both investor behaviour and cryptocurrencies' ecosystem-specific factors. Our results offer valuable implications for market participants and policymakers, helping to inform strategies for managing risk and ensuring market stability.

Future research could extend this framework to incorporate a broader array of cryptocurrencies and explore how emerging developments, such as increased regulatory scrutiny or technological innovation, impact the volatility dynamics uncovered in this study.

Declaration on funding sources

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

Publicly available data can be accessed through the sources referenced in the manuscript. Data requiring a subscription or subject to licensing agreements cannot be shared.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Aas, K., Jullum, M., and Løland, A., 2021. Explaining individual predictions when features are dependent: More accurate approximations to Shapley values. *Artificial Intelligence*, 298, 103502.
- Aboze, B.J., 2023. A comprehensive guide into Shap (Shapley additive explanations) values [online]. *Deepchecks.com*. Available from: <https://deepchecks.com/a-comprehensive-guide-into-shap-shapley-additive-explanations-values/> [Accessed 21 Apr 2024].
- Abreu, D. and Brunnermeier, M.K., 2002. Synchronization risk and delayed arbitrage. *Journal of Financial Economics*, 66 (2–3), 341–360.
- Akepanidtaworn, K., Mascio, R.D., Imas, A., and Schmidt, L.D.W., 2023. Selling fast and buying slow: Heuristics and trading performance of institutional investors. *The Journal of Finance*, 78 (6), 3055–3098.
- Akyildirim, E., Aysan, A.F., Cepni, O., and Serbest, Ö., 2024. Sentiment matters: The effect of news-media on spillovers among cryptocurrency returns. *The European Journal of Finance*, 30 (14), 1577–1613.
- Al Guindy, M., 2021. Cryptocurrency price volatility and investor attention. *International Review of Economics & Finance*, 76, 556–570.
- Andrei, D. and Hasler, M., 2015. Investor attention and stock market volatility. *The Review of Financial Studies*, 28 (1), 33–72.
- Athey, S., Tibshirani, J., and Wager, S., 2019. Generalized random forests. *The Annals of Statistics*, 47 (2).
- Avramov, D., Cheng, S., and Metzker, L., 2023. Machine learning vs. economic restrictions: Evidence from stock return predictability. *Management Science*, 69 (5), 2587–2619.
- Awad, M. and Khanna, R., 2015. Support vector regression. In: *Efficient Learning Machines*. Apress, Berkeley, CA.
- Bai, Y. and Zhang, B., 2024. Fundamental analysis of initial coin offerings. *International Journal of Finance & Economics*, 1–14.
- Banerjee, A.V., 1992. A simple model of herd behavior. *The Quarterly Journal of Economics*, 107 (3), 797–817.
- Barber, B.M. and Odean, T., 2000. Trading is hazardous to your wealth: The common stock investment performance of individual investors. *The Journal of Finance*, 55 (2), 773–806.
- Barber, B.M. and Odean, T., 2008. All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors. *The Review of Financial Studies*, 21 (2), 785–818.
- Barberis, N., Shleifer, A., and Vishny, R., 1998. A model of investor sentiment. *Journal of Financial Economics*, 49 (3), 307–343.
- Baur, D.G. and Dimpfl, T., 2018. Asymmetric volatility in cryptocurrencies. *Economics Letters*, 173, 148–151.
- Ben-David, I., Franzoni, F., Moussawi, R., and Sedunov, J., 2021. The granular nature of large institutional investors. *Management Science*, 67 (11), 6629–6659.
- Ben-Rephael, A., Da, Z., and Israelsen, R.D., 2017. It depends on where you search: Institutional investor attention and underreaction to news. *The Review of Financial Studies*, 30 (9), 3009–3047.
- Bennett, J.A., Sias, R.W., and Starks, L.T., 2003. Greener Pastures and the impact of dynamic institutional preferences. *The Review of Financial Studies*, 16 (4), 1203–1238.
- Biais, B., Bisière, C., Bouvard, M., Casamatta, C., and Menkveld, A.J., 2023. Equilibrium bitcoin pricing. *The Journal of Finance*, 78 (2), 967–1014.

- Białkowski, J., 2020. Cryptocurrencies in institutional investors' portfolios: Evidence from industry stop-loss rules. *Economics Letters*, 191, 108834.
- Bianchi, D., Büchner, M., and Tamoni, A., 2021. Bond risk premiums with machine learning. *The Review of Financial Studies*, 34 (2), 1046–1089.
- Bikhchandani, S., Hirshleifer, D., and Welch, I., 1992. A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100 (5), 992–1026.
- Blankespoor, E., de Haan, E., and Marinovic, I., 2020. Disclosure processing costs, investors' information choice, and equity market outcomes: A Review. *Journal of Accounting and Economics*, 70 (2–3), 101344.
- Bordalo, P., Gennaioli, N., and Shleifer, A., 2013. Saliency and asset prices. *American Economic Review*, 103 (3), 623–628.
- Breiman, L., 2001. Random Forests, *Machine Learning*, 45, 5–32.
- Brunnermeier, M. K. and Nagel, S., 2004. Hedge funds and the technology bubble. *The Journal of Finance*, 59 (5), 2013–2040.
- Cai, F., Han, S., Li, D., and Li, Y., 2019. Institutional Herding and its price impact: Evidence from the Corporate Bond Market. *Journal of Financial Economics*, 131 (1), 139–167.
- Cheah, E. and Fry, J., 2015. Speculative bubbles in Bitcoin markets? An empirical investigation into the fundamental value of Bitcoin. *Economics Letters*, 130, 32–36.
- Chen, S., Härdle, W. K., and Jeong, K., 2010. Forecasting volatility with support vector machine-based GARCH model. *Journal of Forecasting*, 29 (4), 406–433.
- Cheon, Y. and Lee, K., 2017. Maxing Out Globally: Individualism, Investor Attention, and the Cross Section of Expected Stock Returns. *Management Science*, 64 (12), 5807–5831.
- Chiu, P.-C., Lourie, B., Nekrasov, A., and Teoh, S.H., 2021. Cater to thy client: Analyst responsiveness to institutional investor attention. *Management Science*, 67 (12), 7455–7471.
- Choy, S. K. and Wei, J., 2022. Investor attention and option returns. *Management Science*, 69 (8), 4845–4863.
- Coinmetrics.io, 2024. Various, including Crypto-data, Data Encyclopedia, and Data Encyclopedia – Realised Volatility [online]. *Coinmetrics.io*. Available from: <https://charts.coinmetrics.io/crypto-data/> and <https://docs.coinmetrics.io/> [Accessed 24 Feb 2024 – 21 April 2024].
- Cong, L.W., He, Z., and Li, J., 2021b. Decentralized mining in centralized pools. *The Review of Financial Studies*, 34 (3), 1191–1235.
- Cong, L.W., Li, Y., and Wang, N., 2021a. Tokenomics: Dynamic adoption and valuation. *The Review of Financial Studies*, 34 (3), 1105–1155.
- Conlisk, J., 1996. Why Bounded Rationality? *Journal of Economic Literature*, 34 (2), 669–700.
- Cosemans, M. and Frehen, R., 2021. Saliency theory and stock prices: Empirical evidence. *Journal of Financial Economics*, 140 (2), 460–483.
- D'Amato, V., Levantesi, S., and Piscopo, G., 2022. Deep learning in predicting cryptocurrency volatility. *Physica A: Statistical Mechanics and its Applications*, 596, 127158.
- Da, Z., Engelberg, J., and Gao, P., 2011. In search of attention. *The Journal of Finance*, 66 (5), 1461–1499.
- Daniel, K., Hirshleifer, D., and Subrahmanyam, A., 1998. Investor psychology and security market under- and overreactions. *The Journal of Finance*, 53 (6), 1839–1885.
- De Long, J.B., Shleifer, A., Summers, L.H., And Waldmann, R.J., 1990. Positive feedback investment strategies and destabilizing rational speculation. *The Journal of Finance*, 45 (2), 379–395.

- Del Guercio, D., 1996. The distorting effect of the prudent-man laws on Institutional Equity Investments. *Journal of Financial Economics*, 40 (1), 31–62.
- Dellavigna, S. and Pollet, J. M., 2009. Investor Inattention and Friday Earnings Announcements. *The Journal of Finance*, 64(2), 709–749.
- Dennis, P.J. and Strickland, D., 2002. Who blinks in volatile markets, individuals or institutions? *The Journal of Finance*, 57 (5), 1923–1949.
- Dong, S., Fang, V.W., and Lin, W., 2023. Tracing contagion risk: From crypto or stock? Working paper. Available at SSRN: <http://dx.doi.org/10.2139/ssrn.4371403>.
- Duan, K., Keerthi, S. S., and Poo, A. N., 2003. Evaluation of simple performance measures for tuning SVM hyperparameters. *Neurocomputing*, 51, 41–59.
- European Securities and Markets Authority (ESMA), 2024. Markets in crypto-assets regulation (MICA). Available from: <https://www.esma.europa.eu/esmas-activities/digital-finance-and-innovation/markets-crypto-assets-regulation-mica> [Accessed 15 Nov 2024].
- Falkenstein, E. G., 1996. Preferences for stock characteristics as revealed by mutual fund portfolio holdings. *The Journal of Finance*, 51 (1), 111–135.
- Fischer, P.E. and Verrecchia, R.E., 1999. Public information and heuristic trade. *Journal of Accounting and Economics*, 27 (1), 89–124.
- Francq, C. and Thieu, L.Q., 2019. QML inference for volatility models with covariates. *Econometric Theory*, 35 (1), 37–72.
- FRBSL, 2024. Economic Policy Uncertainty Index for United States [USEPUINDXD] [online]. *Federal Reserve Bank of St. Louis*. Based on Baker, S., Bloom, N. and Davis, S., 2015. Measuring Economic Policy Uncertainty. Working Paper 21633. NBER Working Paper Series. Available from: <https://fred.stlouisfed.org/series/USEPUINDXD> [Accessed 25 Feb 2024].
- Froot, K.A., Scharfstein, D.S., and Stein, J.C., 1992. Herd on the street: Informational inefficiencies in a market with short-term speculation. *The Journal of Finance*, 47 (4), 1461–1484.
- Fry, J. 2018. Booms, busts and heavy-tails: The story of Bitcoin and cryptocurrency markets? *Economics Letters*, 171, 225–229.
- Gandal, N., Hamrick, J., Moore, T., and Oberman, T., 2018. Price manipulation in the Bitcoin ecosystem, *Journal of Monetary Economics*, 95, 86–96.
- Gargano, A. and Rossi, A. G., 2018. Does it pay to pay attention? *The Review of Financial Studies*, 31 (12), 4595–4649.
- Gnecco, G., Nutarelli, F., and Riccaboni, M., 2023. Matrix completion of World Trade: An analysis of interpretability through Shapley Values. *The World Economy*, 46 (9), 2707–2731.
- Gompers, P. A. and Metrick, A., 2001. Institutional investors and equity prices. *The Quarterly Journal of Economics*, 116 (1), 229–259.
- Griffin, J. M., Harris, J. H., and Topaloglu, S., 2003. The dynamics of institutional and individual trading. *The Journal of Finance*, 58 (6), 2285–2320.
- Griffin, J.M. and Shams, A., 2020. Is bitcoin really untethered? *The Journal of Finance*, 75 (4), 1913–1964.
- Grossman, S. J. and Stiglitz, J. E., 1980. On the Impossibility of Informationally Efficient Markets. *The American Economic Review*, 70 (3), 393–408.
- Gu, S., Kelly, B., and Xiu, D., 2020. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33 (5), 2223–2273.
- Hackethal, A., Hanspal, T., Lammer, D. M., and Rink, K., 2022. The characteristics and portfolio behavior of bitcoin investors: Evidence from indirect cryptocurrency investments. *Review of Finance*, 26 (4), 855–898.

- Han, H. and Kristensen, D., 2014. Asymptotic theory for the QMLE in GARCH-X models with stationary and nonstationary covariates. *Journal of Business & Economic Statistics*, 32 (3), 416–429.
- Haykin, S.S., 1999. *Neural Networks: a comprehensive foundation*. 2nd ed. Englewood Cliffs, N.J.: Prentice Hall.
- Hirshleifer, D., Subrahmanyam, A., and Titman, S., 1994. Security analysis and trading patterns when some investors receive information before others. *The Journal of Finance*, 49 (5), 1665–1698.
- Hvidkjaer, S., 2008. Small trades and the cross-section of Stock returns. *The Review of Financial Studies*, 21 (3), 1123–1151.
- Iliev, P., Kalodimos, J., and Lowry, M., 2021. Investors' attention to corporate governance. *The Review of Financial Studies*, 34 (12), 5581–5628.
- James, G., Witten, D., Hastie, T., and Tibshirani, R., 2017. *An introduction to statistical learning with applications in R*. New York: Springer.
- Jang, J. and Kang, J., 2019. Probability of price crashes, rational speculative bubbles, and the cross-section of Stock returns. *Journal of Financial Economics*, 132 (1), 222–247.
- Jermann, U.J. 2021. Cryptocurrencies and Cagan's model of hyperinflation, *Journal of Macroeconomics*, 69, 103340.
- Kahneman, D. and Tversky, A., 1973. On the psychology of prediction. *Psychological Review*, 80 (4), 237–251.
- Kahneman, D., 1973. *Attention and effort*. Englewood Cliffs, N. J, Prentice-Hall.
- Khashanah, K. and Shao, C., 2022. Short-term volatility forecasting with kernel support vector regression and Markov switching multifractal model. *Quantitative Finance*, 22 (2), 241–253.
- Kogan, S., Makarov, I., Niessner, M. and Schoar, A. 2024. Are crypto different? Evidence from retail trading. *Journal of Financial Economics*, 159, 103897.
- Kyle, A. S. and Wang, F. A., 1997. Speculation duopoly with agreement to disagree: Can overconfidence survive the market test? *The Journal of Finance*, 52 (5), 2073–2090.
- Leippold, M., Wang, Q., and Zhou, W., 2022. Machine learning in the Chinese Stock Market. *Journal of Financial Economics*, 145 (2), 64–82.
- Lin, T., Horne, B.G., Tino, P., and Giles, C.L., 1996. Learning long-term dependencies in NARX Recurrent Neural Networks. *IEEE Transactions on Neural Networks*, 7 (6), 1329–1338.
- Liu, A., Jahanshahloo, H., Chen, J., and Eshraghi, A., 2023. Trading patterns in the Bitcoin market. *The European Journal of Finance*, 1–18.
- Liu, Y. and Tsyvinski, A., 2021. Risks and returns of cryptocurrency. *The Review of Financial Studies*, 34 (6), 2689–2727.
- Liu, Y., Tsyvinski, A., and Wu, X., 2022. Common risk factors in cryptocurrency. *The Journal of Finance*, 77 (2), 1133–1177.
- Long, S., Chatziantoniou, I. Gabauer, D. and Lucey, B. 2024. Do social media sentiments drive cryptocurrency intraday price volatility? New evidence from asymmetric TVP-VAR frequency connectedness measures, *The European Journal of Finance*, 30 (13), 1470–1489.
- Lundberg, S. M. and Lee, S., 2017. A unified approach to interpreting model predictions. *In Proceedings of the 31st international conference on neural information processing systems*, 4768–4777.
- Makarov, I., and A. Schoar. 2020. Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics* 135:293–319.
- Makarov, I., and A. Schoar. 2022. Blockchain Analysis of the Bitcoin Market. *Working paper, NBER, LSE and MIT*.

- MathWorks, 2024. Various, including documentation for “Deep Learning” and “Statistics and Machine Learning” Toolboxes, MATLAB [online], *MathWorks.com*. [Accessed 10 Mar 2024 - 21 Apr 2024].
- Menezes, J. M. and Barreto, G. A., 2008. Long-term time series prediction with the NARX network: An empirical evaluation. *Neurocomputing*, 71 (16–18), 3335–3343.
- Merkley, K.J., Pacelli, J., Piorkowski, M., and Williams, B., 2024. Crypto influencers. *Review of Accounting Studies*, 29 (3), 2254–2297.
- Molnar, C., 2023. Interpretable machine learning: A Guide for Making Black Box Models Explainable [online]. Leanpub. Available from: <https://christophm.github.io/interpretable-ml-book/> [Accessed 21 Apr 2024].
- Murphy, K.P., 2012. Machine learning: A probabilistic perspective. Cambridge, Massachusetts: The MIT Press.
- Nofsinger, J. R. and Sias, R. W. 1999. Herding and Feedback Trading by Institutional and Individual Investors, *The Journal of Finance*, 54 (6), 2263–2295.
- Ozdamar, M., Sensoy, A., and Akdeniz, L., 2022. Retail vs institutional investor attention in the cryptocurrency market. *Journal of International Financial Markets, Institutions and Money*, 81 (101674).
- Pagnotta, E., 2022. Decentralizing Money: Bitcoin Prices and Blockchain Security. *The Review of Financial Studies*, 35(2), 866–907.
- Palomino, F., 1996. Noise trading in small markets. *The Journal of Finance*, 51 (4), 1537–1550.
- Pattnaik, D., Hassan, M. K., Dsouza, A., Tiwari, A., and Devji, S., 2023. Ex-post facto analysis of cryptocurrency literature over a decade using bibliometric technique. *Technological Forecasting and Social Change*, 189, 122339.
- Pedersen, R.S. and Rahbek, A., 2019. Testing GARCH-X type models. *Econometric Theory*, 35 (5), 1012–1047.
- Peng, L. and Xiong, W., 2006. Investor attention, overconfidence and category learning. *Journal of Financial Economics*, 80 (3), 563–602.
- Peng, Y., Albuquerque, P.H., Camboim de Sá, J.M., Padula, A.J., and Montenegro, M.R., 2018. The best of two worlds: Forecasting high frequency volatility for cryptocurrencies and traditional currencies with support vector regression. *Expert Systems with Applications*, 97, 177–192.
- Ruiz, L., Cuéllar, M., Calvo-Flores, M. and Jiménez, M., 2016. An Application of Non-Linear Autoregressive Neural Networks to Predict Energy Consumption in Public Buildings. *Energies*, 9(9), 684.
- Scharfstein, D. S. and Stein, J. C., 1990. Herd behavior and investment. *American Economic Review*, 80 (3), 465–479.
- Schilling, L., and H. Uhlig. 2019. Some simple bitcoin economics. *Journal of Monetary Economics* 106:16–26.
- Schölkopf, B. and Smola, A. J., 2001. Learning with kernels: Support vector machines, regularization, optimization and beyond. Cambridge, MA: MIT press.
- Shen, D., Urquhart, A., and Wang, P., 2019. Does Twitter predict bitcoin? *Economics Letters*, 174, 118–122.
- Shleifer, A. and Vishny, R. W., 1997. The limits of arbitrage. *The Journal of Finance*, 52 (1), 35–55.
- Sias, R. W. and Whidbee, D.A., 2010. Insider trades and demand by institutional and individual investors. *The Review of Financial Studies*, 23 (4), 1544–1595.
- Sias, R. W., 2004. Institutional herding. *The Review of Financial Studies*, 17 (1), 165–206.
- Sicherman, N., Loewenstein, G., Seppi, D.J., and Utkus, S.P., 2016. Financial attention. *The Review of Financial Studies*, 29 (4), 863–897.

- Smales, L. A., 2022. Investor attention in cryptocurrency markets. *International Review of Financial Analysis*, 79, 101972.
- Sockin, M. and Xiong, W., 2023. A model of cryptocurrencies. *Management Science*, 69(11), 6684-6707.
- Stasinakis, C., Sermpinis, G., Psaradellis, I., and Verousis, T., 2016. Krill-herd support vector regression and heterogeneous autoregressive leverage: Evidence from forecasting and trading commodities. *Quantitative Finance*, 16 (12), 1901–1915.
- Trueman, B., 1988. A theory of noise trading in securities markets. *The Journal of Finance*, 43 (1), 83–95.
- Trueman, B., 1994. Analyst forecasts and herding behavior: *The Review of Financial Studies*, 7 (1), 97–124.
- Umar, M., Rizvi, S. K., and Naqvi, B., 2021. Dance with the devil? the nexus of fourth industrial revolution, technological financial products and volatility spillovers in Global Financial System. *Technological Forecasting and Social Change*, 163, 120450.
- Vapnik, V., Golowich, S. E., and Smola, A., 1997. Support vector method for function approximation, regression estimation, and signal processing. *Advances in Neural Information Processing Systems*, 281–287.
- Verrecchia, R. E., 2001. Essays on disclosure. *Journal of Accounting and Economics*, 32 (1–3), 97–180.
- Wagner, S., and Athey, S., 2018. Estimation and Inference of Heterogeneous Treatment Effectst using Random Forests, *Journal of the American Statistical Association*, 113:523, 1228-1242.
- Walther, T., Klein, T., and Bouri, E., 2019. Exogenous drivers of Bitcoin and cryptocurrency volatility – a mixed data sampling approach to forecasting. *Journal of International Financial Markets, Institutions and Money*, 63, 101133.
- Wang, Y., Andreeva, G., and Martin-Barragan, B., 2023. Machine learning approaches to forecasting cryptocurrency volatility: Considering internal and external determinants. *International Review of Financial Analysis*, 90, 102914.
- White House, 2022. Executive order on ensuring responsible development of Digital assets [online]. The United States Government. Available from: <https://www.whitehouse.gov/briefing-room/presidential-actions/2022/03/09/executive-order-on-ensuring-responsible-development-of-digital-assets/> [Accessed 15 Nov 2024].
- Zhang, C., Ma, H., Arkorful, G. B., and Peng, Z., 2023. The impacts of futures trading on volatility and volatility asymmetry of Bitcoin returns. *International Review of Financial Analysis*, 86, 102497.

Table 1 Descriptive statistics

	(1)	(2)	(3)
Daily average	BTC	ETH	XRP
Closing price (USD)	12,348.6091 (15,787.8470)	1,177.6299 (1,166.9239)	0.5178 (0.3402)
Continuously compounded returns	0.0011 (0.0386)	0.0005 (0.0503)	-0.0002 (0.0593)
Realised volatility	0.0357 (0.0160)	0.0474 (0.0176)	0.0533 (0.0299)
Individual investor attention (Twitter count, ln)	9.3167 (1.4926)	10.0342 (1.3838)	9.2186 (0.8565)
Individual investor attention (Twitter count, raw)	31,964.9953 (45,481.5538)	55,659.7852 (69,546.0694)	15,123.3289 (17,631.3050)
Institutional investor attention (Bloomberg news count, ln)	5.0698 (1.4725)	4.4226 (1.0233)	3.8930 (0.5212)
Institutional investor attention (Bloomberg news count, raw)	324.2837 (343.4171)	131.6539 (133.7727)	54.1619 (27.5320)
Observations	3412	1965	1964
Sample start date	01.01.2014	18.12.2017	19.12.2017

Note: Table reports the daily mean and standard deviation (in parentheses) of the variables. Investor attention variables are measured by the natural logarithm of one plus the daily count of Tweets or Bloomberg news for each cryptocurrency.

Table 2: Cryptocurrency holdings

	(1)	(2)	(3)
Panel A	BTC	ETH	XRP
Average number of addresses with a balance greater than:			
0.1% of supply	53 (10)	128 (21)	196 (28)
0.01% of supply	956 (72)	1,040 (83)	401 (71)
1 million units	0 (0)	6 (1)	1677 (178)
100K units	3 (1)	154 (27)	18,384 (6884)
10K units	106 (11)	1,242 (60)	127,694 (58,750)
1K units	1,896 (250)	7,541 (652)	418,360 (172,891)
Panel B			
Total supply, units	16,820,083 (2,007,553)	111,240,312 (6,999,020)	99,990,698,374 (1,023,711)
Supply held by top 1% of addresses, units	14,823,367 (2,491,264)	108,288,927 (7,124,350)	96,373,675,718 (1,551,236,138)
<i>% of total supply</i>	87.6 (5.4)	97.3 (0.5)	96.4 (1.6)
Supply held by top 10% of addresses, units	16,660,356 (1,968,021)	110,931,763 (6,933,883)	99,541,104,468 (1,689,206,82)
<i>% of total supply</i>	99.1 (0.2)	99.7 (0.1)	99.6 (0.2)
Supply held by top 100 addresses, units	2,855,380 (201,894)	40,417,640 (5,195,058)	63,427,770,829 (7,854,765,859)
<i>% of total supply</i>	17.2 (2.3)	36.2 (2.8)	63.4 (7.9)
Observations	3412	1965	1964

Note: Table reports the mean and standard deviation (in parentheses) values of the variables. Data frequency is daily.

Table 3: Modelling cryptocurrency volatility and investor attention using GARCH-X family models

	(1)	(2)	(3)
	BTC	ETH	XRP
Dependent variable: Continuously compounded returns	TGARCH (3,1,2)	GJR-GARCH (2,1,2)	GJR-GARCH (3,1,2)
Investor attention			
Individual investor attention	0.1474 (0.6233)	0.1614 (0.2068)	0.0705 (0.9027)
Institutional investor attention	1.7888**** (0.0000)	0.7630**** (0.0000)	1.7801** (0.0276)
Constant	-21.8390**** (0.0000)	-6.7644**** (0.0000)	-6.9978* (0.0942)
Portmanteau Q test - Standardised residuals			
P-value (Q test)	0.2733	0.7099	0.9514
Q statistics	44.9198	34.6388	26.4147
Portmanteau Q test - Squared standardised residuals			
P-value (Q test)	0.2260	0.2453	0.9541
Q statistics	46.3776	45.7624	26.2322
Accuracy measures			
Root mean square error	0.0164	0.0179	0.0226
Mean absolute error	0.0102	0.0121	0.0155
Observations	3412	1964	1964

p-values in parentheses * $p < .10$, ** $p < .05$, *** $p < .01$, **** $p < .001$

Note: Investor attention variables are measured by the natural logarithm of one plus the daily count of Tweets or Bloomberg news for each cryptocurrency. Several variables are differenced to achieve stationarity. Models use the robust or sandwich estimator of variance. This estimator is robust to certain types of misspecifications. Time effect is captured using quarterly time dummies in all models. In asymmetric GARCH-X models, values in brackets (p, r, q) stand for: p represents the lags of the GARCH component, r is the leverage parameter, and q is the lags of the ARCH component, respectively (also for Tables 4 and 5).

Table 4: GARCH-X family models with the whale effect ($\geq 0.1\%$ of supply) and control variables

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	BTC		ETH		XRP	
Continuously compounded returns	GJR-GARCH (3,2,3)	GJR-GARCH (2,2,2)	ARIMA (1,0,0) - TGARCH (1,3,3)	ARIMA (0,0,2) - TGARCH (3,3,3)	ARIMA (0,0,1) - TGARCH (2,3,2)	ARIMA (1,0,0) - TGARCH (2,2,2)
<i>Investor attention</i>						
Individual investor attention	0.4943* (0.0707)	0.6266* (0.0820)	0.0308 (0.9212)	0.0092 (0.8413)	-0.4824*** (0.0012)	-0.3587*** (0.0046)
Institutional investor attention	0.5064**** (0.0004)	0.7395**** (0.0000)	0.6379**** (0.0009)	0.6693**** (0.0001)	0.6237**** (0.0000)	0.2998**** (0.0000)
<i>Cryptocurrency holdings and interaction</i>						
Addresses with balance $\geq 0.1\%$ of supply, count	2.1512 (0.5535)	0.0356 (0.3955)	12.0895** (0.0387)	0.1313** (0.0319)	6.1089 (0.5245)	0.0120 (0.8041)
Institutional investor attention $\times 0.1\%$ of supply		-0.0326 (0.3784)		-0.0109 (0.8245)		-0.0149 (0.6728)
<i>Network factors</i>						
Active addresses count	2.2156*** (0.0014)	0.9333*** (0.0014)	-0.6404 (0.2503)	-0.2467 (0.1861)	1.0428**** (0.0000)	1.1544**** (0.0000)
Mean transaction fee (USD)	1.4952*** (0.0023)	0.2840** (0.0129)	1.3010**** (0.0006)	0.3592*** (0.0069)	0.5346**** (0.0001)	0.6086**** (0.0001)
Transaction count	1.0775** (0.0409)	0.5870*** (0.0090)	3.0007*** (0.0012)	1.0109**** (0.0002)	1.9449**** (0.0000)	0.8011**** (0.0000)
<i>Financial market factors</i>						
VIX	0.8385*** (0.0011)	0.3148**** (0.0000)	0.7467**** (0.0006)	0.2621**** (0.0004)	0.9615**** (0.0001)	0.3121**** (0.0001)
S&P 500 index	-19.6563**** (0.0001)	-0.1510**** (0.0007)	-3.6269 (0.5027)	-0.0453 (0.4452)	-4.2205 (0.2480)	-0.0499 (0.2313)
<i>Cryptocurrency specific factor</i>						
Market capitalisation	-0.8197** (0.0145)	-1.3635** (0.0114)	-2.1107** (0.0482)	-0.1033* (0.0732)	-0.3312 (0.3151)	-0.2124 (0.2553)
<i>Uncertainty factor</i>						
EPUI US	-0.0963 (0.4471)	-0.0137 (0.8533)	-0.0986 (0.3277)	-0.0547 (0.4015)	-0.1429 (0.2091)	-0.0774 (0.2530)
Constant	-39.8036**** (0.0000)	-2.4402*** (0.0021)	-41.3045**** (0.0000)	-2.7960**** (0.0000)	-34.4661**** (0.0001)	-9.4469**** (0.0000)
Portmanteau Q test - Standardised residuals						
P-value (Q test)	0.3209	0.2646	0.6178	0.6697	0.9026	0.9027
Q statistics	43.6020	45.1742	36.7408	35.5725	28.9445	28.9397
Portmanteau Q test - Squared standardised residuals						
P-value (Q test)	0.1930	0.0625	0.1739	0.1299	0.1128	0.0828
Q statistics	47.5211	54.5330	48.2481	50.1785	51.0652	52.9283
Accuracy measures						
Root mean square error	0.0160	0.0162	0.0222	0.0216	0.0302	0.0296
Mean absolute error	0.0111	0.0114	0.0140	0.0140	0.0197	0.0195

Observations	3412	3412	1964	1964	1964	1964
--------------	------	------	------	------	------	------

p-values in parentheses * $p < .10$, ** $p < .05$, *** $p < .01$, **** $p < .001$

Note (also for Table 5 below): EPUI US stands for economic policy uncertainty index in the US, and VIX for volatility index. Investor attention variables are measured by the natural logarithm of one plus the daily count of Tweets or Bloomberg news for each cryptocurrency. All the other variables have undergone standard logarithmic transformation. All variables in models 2,4, and 6 are standardised. Several variables are differenced to achieve stationarity. Models use the robust or sandwich estimator of variance. This estimator is robust to certain types of misspecifications. Time effect is captured using quarterly time dummies in all models.

Table 5: GARCH-X family models with the whale effect ($\geq 0.01\%$ of supply) and control variables

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	BTC		ETH		XRP	
Continuously compounded returns	GJR-GARCH (3,2,2)	GJR-GARCH (2,3,1)	ARIMA (1,0,0) - TGARCH (2,3,3)	ARIMA (0,0,2) - TGARCH (3,2,2)	ARIMA (0,0,1) - TGARCH (2,1,3)	TGARCH (2,2,2)
<i>Investor attention</i>						
Individual investor attention	0.4997* (0.0753)	0.7755** (0.0326)	0.1410 (0.6488)	0.0630 (0.3386)	-0.3930*** (0.0065)	-0.3217** (0.0119)
Institutional investor attention	0.5039**** (0.0006)	0.7612**** (0.0000)	0.7033**** (0.0000)	0.7734**** (0.0000)	0.5568**** (0.0000)	0.2821**** (0.0000)
<i>Cryptocurrency holdings and interaction</i>						
Addresses with balance $\geq 0.01\%$ of supply, count	5.3899** (0.0345)	0.2626 (0.1645)	-14.8663 (0.1934)	-0.0437 (0.3259)	-3.8535 (0.5587)	-0.0291 (0.4840)
Institutional investor attention $\times 0.01\%$ of supply		-0.1592 (0.1598)		0.0140 (0.6705)		0.0312 (0.2822)
<i>Network factors</i>						
Active addresses count	2.2932*** (0.0011)	0.9831**** (0.0008)	-0.6406 (0.2200)	-0.3429 (0.2801)	1.0214**** (0.0000)	1.0639**** (0.0000)
Mean transaction fee (USD)	1.6629**** (0.0002)	0.3228** (0.0197)	1.1031**** (0.0000)	0.2467**** (0.0005)	0.5692**** (0.0001)	0.6965**** (0.0002)
Transaction count	0.9855* (0.0625)	0.5068** (0.0220)	3.2743**** (0.0000)	1.1510**** (0.0000)	2.0430**** (0.0000)	0.7955**** (0.0000)
<i>Financial market factors</i>						
VIX	0.7821*** (0.0021)	0.2962**** (0.0001)	0.7493*** (0.0010)	0.2215*** (0.0052)	0.9014**** (0.0001)	0.3183**** (0.0000)
S&P 500 index	-20.0827**** (0.0000)	-0.1457*** (0.0010)	-2.3618 (0.6091)	-0.0848** (0.0266)	-4.8719 (0.1894)	-0.0450 (0.2639)
<i>Cryptocurrency specific factor</i>						
Market capitalisation	-0.7637** (0.0274)	-1.4387** (0.0213)	-1.6472** (0.0486)	-0.0618 (0.1334)	-0.5112 (0.1193)	-0.2283 (0.2132)
<i>Uncertainty factor</i>						
EPUI US	-0.0961 (0.4504)	-0.0395 (0.5928)	-0.1229 (0.2203)	-0.0409 (0.6987)	-0.1032 (0.3533)	-0.0639 (0.3455)
Constant	-78.7005**** (0.0002)	-3.2979**** (0.0000)	-44.9688**** (0.0000)	-3.0099**** (0.0000)	-31.5275**** (0.0003)	0.8299 (0.1945)
Portmanteau Q test - Standardised residuals						
P-value (Q test)	0.3361	0.2617	0.6269	0.6050	0.8962	0.8085
Q statistics	43.2027	45.2616	36.5370	37.0227	29.2057	32.1066
Portmanteau Q test - Squared standardised residuals						
P-value (Q test)	0.2549	0.0666	0.2182	0.2035	0.2423	0.1500
Q statistics	45.4663	54.1740	46.6391	47.1430	45.8558	49.2417
Accuracy measures						
Root mean square error	0.0162	0.0164	0.0212	0.0208	0.0301	0.0294
Mean absolute error	0.0111	0.0114	0.0140	0.0139	0.0196	0.0192

Observations	3412	3412	1964	1964	1964	1964
<i>p-values in parentheses * $p < .10$, ** $p < .05$, *** $p < .01$, **** $p < .001$</i>						

Table 6: NARX modelling of cryptocurrency volatility and investor attention

	(1)	(2)	(3)
	BTC	ETH	XRP
Accuracy measures	NARX (4-15)	NARX (5-20)	NARX (6-30)
Mean absolute error (MAE)	0.0012	0.0015	0.0019
Root mean square error (RMSE)	0.0025	0.0029	0.0037
Training performance (RMSE)	0.0026	0.0029	0.0037
Validation performance (RMSE)	0.0023	0.0027	0.0036
Test performance (RMSE)	0.0021	0.0026	0.0034

Note (also for Table 7 below): These models use the same independent (predictor) variables as in Table 3. Several variables are differenced to achieve stationarity. All neural networks use one hidden layer. The Levenberg-Marquardt (LM) algorithm is chosen for training across all models. NARX (4-15), NARX (5-20) and NARX (6-30) signify neural networks with 4, 5, and 6 delays/lags, respectively, and 15, 20, and 30 neurons. The first two rows report in-sample accuracy measures of MAE and RMSE (i.e., cover full sample), while the following three rows report RMSE values for each of the training, validation and test samples. Training, validation and test sizes are set as 70%, 15% and 15% of the sample, respectively.

Table 7: NARX modelling of cryptocurrency volatility and investor attention including the whale effect and control variables

	(1)	(2)	(3)	(4)	(5)	(6)
	BTC		ETH		XRP	
Whale effect ($\geq 0.1\%$ of supply)						
Accuracy measures	NARX (2-20)	NARX (2-15)	NARX (6-10)	NARX (6-15)	NARX (2-20)	NARX (2-15)
Mean absolute error (MAE)	0.0013	0.0013	0.0016	0.0017	0.0020	0.0019
Root mean square error (RMSE)	0.0024	0.0023	0.0027	0.0027	0.0035	0.0035
Training performance (RMSE)	0.0024	0.0023	0.0028	0.0027	0.0036	0.0036
Validation performance (RMSE)	0.0023	0.0022	0.0026	0.0026	0.0033	0.0034
Test performance (RMSE)	0.0022	0.0020	0.0024	0.0024	0.0032	0.0033
Whale effect ($\geq 0.01\%$ of supply)						
Accuracy measures	NARX (2-20)	NARX (6-15)	NARX (5-10)	NARX (3-10)	NARX (3-30)	NARX (3-15)
Mean absolute error (MAE)	0.0013	0.0014	0.0017	0.0015	0.0020	0.0020
Root mean square error (RMSE)	0.0024	0.0024	0.0027	0.0027	0.0036	0.0036
Training performance (RMSE)	0.0025	0.0025	0.0027	0.0027	0.0037	0.0037
Validation performance (RMSE)	0.0023	0.0023	0.0026	0.0026	0.0034	0.0035
Test performance (RMSE)	0.0021	0.0022	0.0025	0.0025	0.0032	0.0030

Note: These models use the same independent (predictor) variables as in Tables 4 and 5. The models in columns 2, 4, and 6 include an interaction term between the respective whale effect and institutional investor attention variables.

Figure 1: Realised volatility

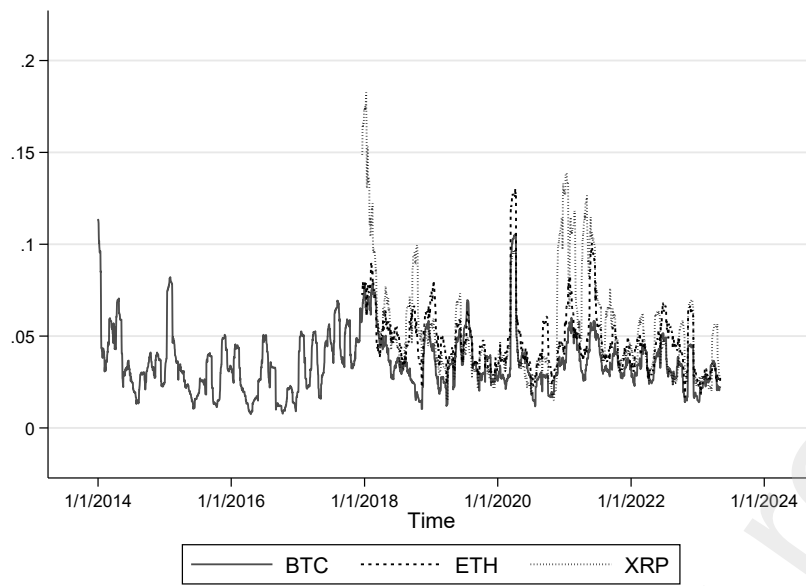


Figure 2: Cryptocurrency holdings

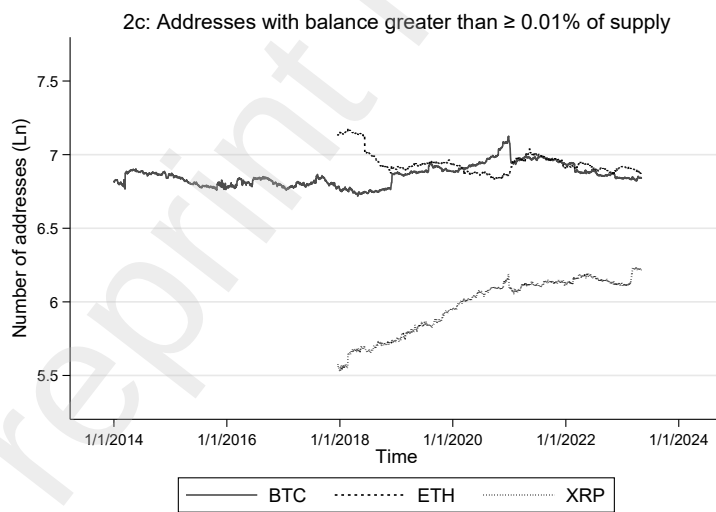
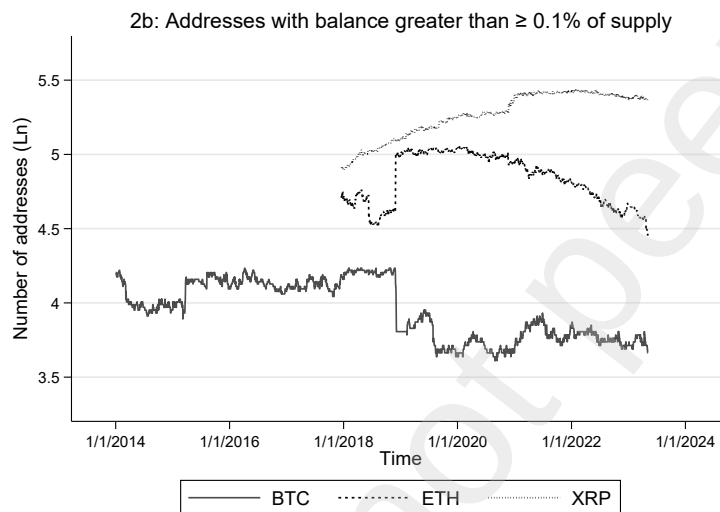
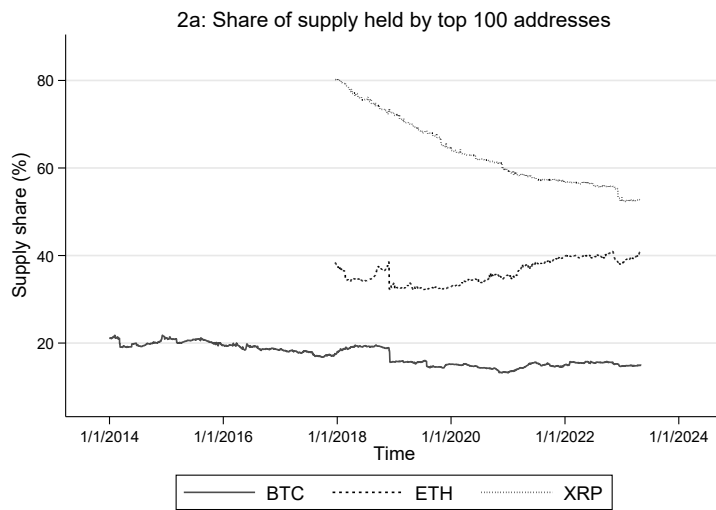


Figure 3: Individual and institutional investor attention

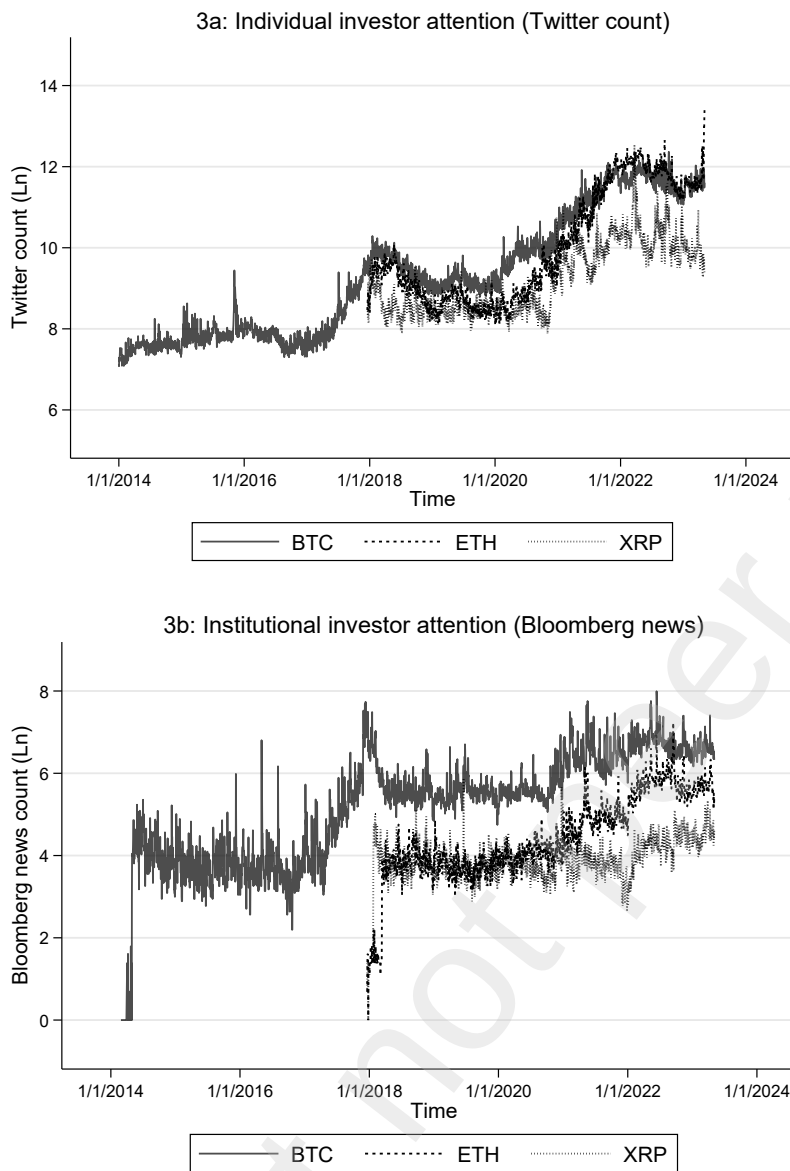


Figure 4: Shapley importance and summary plots from RF with investor attention measures

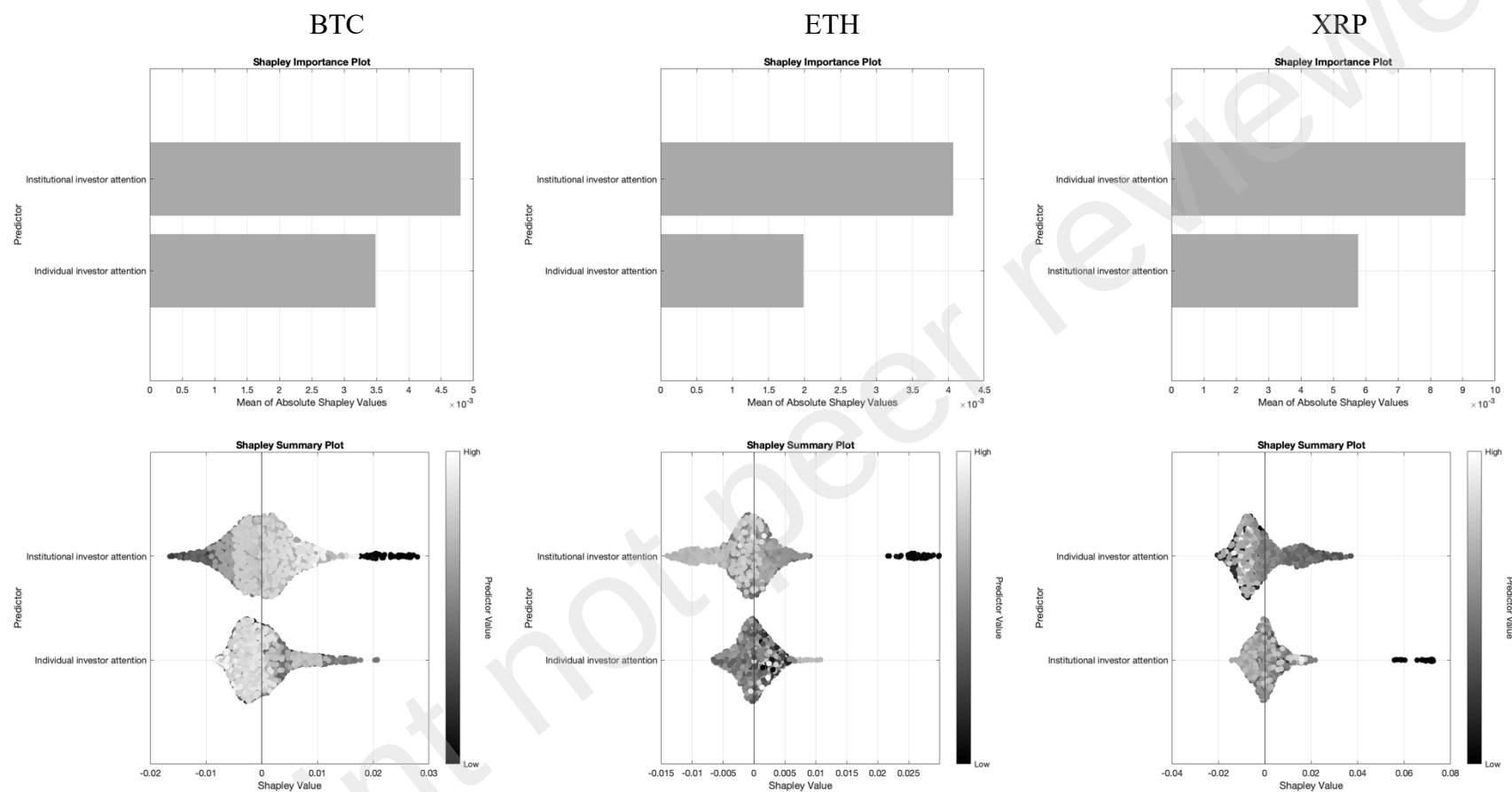
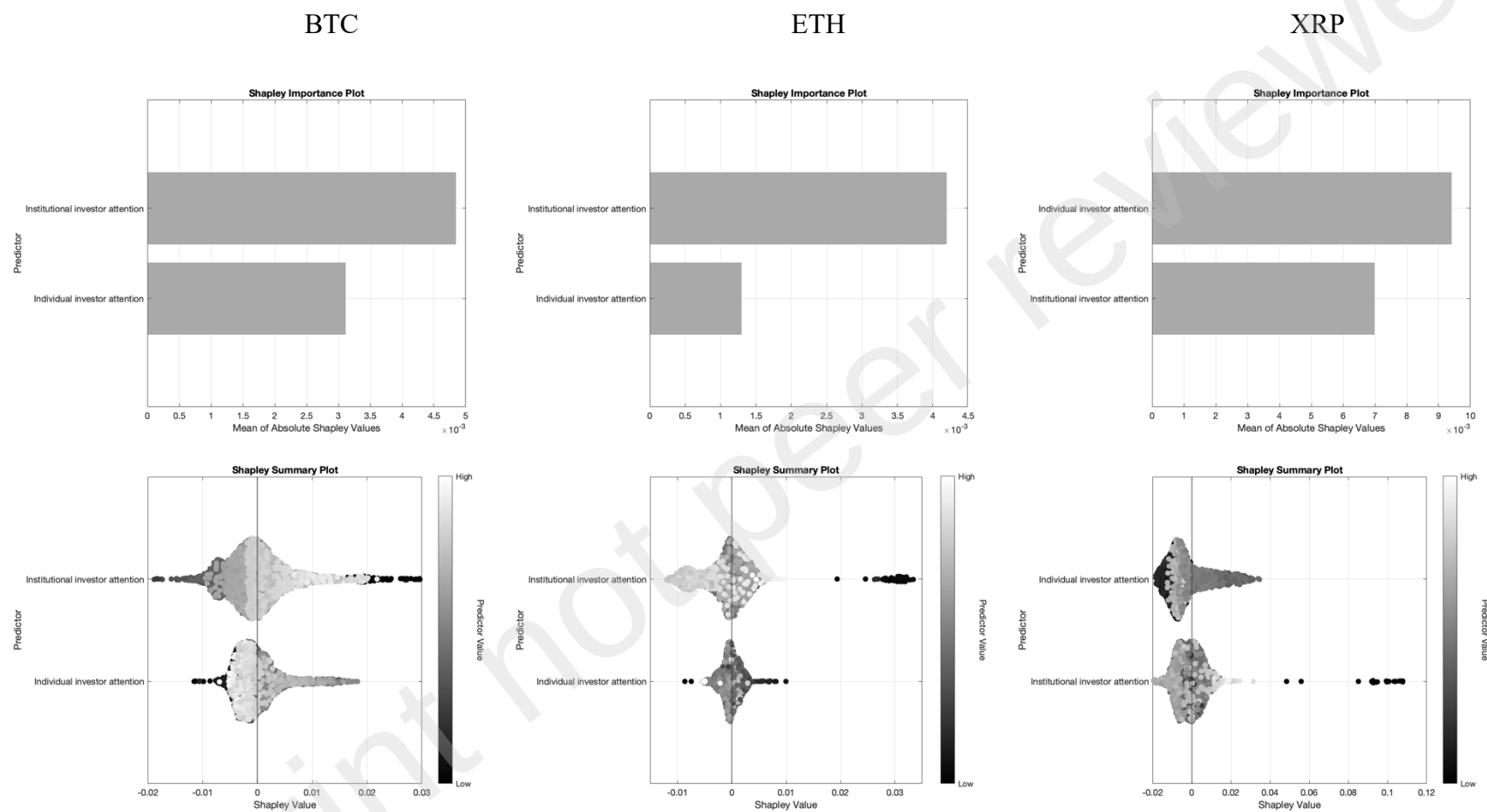


Figure 5: Shapley importance and summary plots from SVR with investor attention measures



Appendix

Table A1: Description of variables

Variables	Description	Sources
<i>Investor attention</i>		
Twitter count	The volume of Twitter (X) data for a specific search query such as BTC, ETH and XRP.	Twitter API v2
Bloomberg news	Bloomberg Terminal story counts (News only) from the securities section for each cryptocurrency. The news trend function was applied.	retrieved from Bloomberg Terminal
<i>Cryptocurrency holdings</i>		
Addresses with balance greater than $\geq 0.1\%$ of supply, count	The sum count of unique addresses holding at least 0.1% ownership of the current supply of native units.	Coinmetrics.io (2024)
Addresses with balance greater than $\geq 0.01\%$ of supply, count	The sum count of unique addresses holding at least 0.01% ownership of the current supply of native units.	
Addresses with balance greater than 1 million units	The sum count of unique addresses holding at least 1,000,000 native units.	
Addresses with balance greater than 100K units	The sum count of unique addresses holding at least 100,000 native units.	
Addresses with balance greater than 10K units	The sum count of unique addresses holding at least 10,000 native units.	
Addresses with balance greater than 1K units	The sum count of unique addresses holding at least 1,000 native units.	
Supply held by top 1% of addresses, units	The sum of all native units held by the richest (top) 1% of addresses.	
Supply held by top 10% of addresses, units	The sum of all native units held by the richest (top) 10% of addresses.	
Supply held by top 100 addresses, units	The sum of all native units held by the richest (top) 100 addresses.	
<i>Network factors</i>		
Active addresses count	The number of unique addresses that are active in the network – used as a proxy for the number of users in a blockchain network.	Coinmetrics.io (2024)
Transaction count	The sum of transactions on a given day.	
Mean transaction fee (USD)	The USD value of the mean fee per transaction.	
<i>Financial market factors</i>		
S&P 500 index	The last price of Standard and Poor’s 500 index.	retrieved from Bloomberg Terminal
VIX	Chicago Board Options Exchange volatility index measures market expectation of near-term volatility conveyed by the S&P 500 index.	
<i>Cryptocurrency specific factors</i>		

Market capitalisation	The sum USD value of the current supply. Also referred to as network value or market capitalisation. Computed as supply $\times$ price. The price used is the daily close price. For XRP, amounts under escrow are included in current supply.	Coinmetrics.io (2024)
<i>Global or Uncertainty factor</i>		
EPUI US	The daily Economic Policy Uncertainty Index is based on newspapers in the United States.	FRBSL (2024), Federal Reserve Bank of St. Louis, Economic Policy Uncertainty Index for United States [USEPUINDXD]. Based on Baker, S., Bloom, N. and Davis, S., 2015. Measuring Economic Policy Uncertainty. Working Paper 21633. NBER Working Paper Series.

Note: Some of the variables have values only on trading days (mainly data obtained from Bloomberg and Federal Reserve Bank of St. Louis). Thus, we replace the closing value of Saturdays and Sundays with those of Fridays.

Table A2: Descriptive statistics of all variables used in the analysis

Variables	BTC		ETH		XRP	
	Mean/SD	Skewness/Kurtosis	Mean/SD	Skewness/Kurtosis	Mean/SD	Skewness/Kurtosis
Continuously compounded returns	0.0011 (0.0386)	-0.6892 (13.2936)	0.0005 (0.0503)	-0.9598 (13.3273)	-0.0002 (0.0593)	0.4107 (15.6112)
Realised volatility	0.0357 (0.0160)	1.2566 (5.8410)	0.0474 (0.0176)	1.7052 (7.7125)	0.0533 (0.0299)	1.6152 (5.5875)
<i>Investor attention</i>						
Individual investor attention	9.3167 (1.4926)	0.3715 (1.8305)	0.0025 (0.1537)	0.2155 (5.0633)	9.2186 (0.8565)	0.5503 (2.3032)
Institutional investor attention	5.0698 (1.4725)	-1.1223 (4.7881)	4.4226 (1.0233)	-0.4281 (3.8347)	3.8930 (0.5212)	-1.4281 (10.1933)
<i>Cryptocurrency holdings</i>						
Addresses with balance greater than $\geq 0.1\%$ of supply, count	-0.0002 (0.0123)	-0.4931 (14.1597)	-0.0001 (0.0107)	4.0855 (130.0148)	0.0002 (0.0048)	1.8300 (17.0738)
Addresses with balance greater than $\geq 0.01\%$ of supply, count	6.8599 (0.0736)	0.5704 (3.0773)	-0.0001 (0.0047)	-7.3140 (190.0856)	0.0003 (0.0064)	0.9220 (17.0707)
<i>Network factors</i>						
Active addresses count	13.2862 (0.5629)	-1.0511 (3.0801)	12.9744 (0.3266)	-0.3029 (2.1815)	9.7226 (1.1458)	0.5889 (2.0813)
Mean transaction fee (USD)	0.0012 (0.2015)	0.4047 (18.2525)	0.0017 (0.3027)	0.0181 (19.2756)	-7.4168 (1.1646)	0.5064 (3.1582)
Transaction count	12.2639 (0.5068)	-1.2649 (3.6049)	13.7113 (0.3108)	-0.4702 (2.0374)	13.7914 (0.3967)	-0.3559 (3.7735)
<i>Financial market factors</i>						
VIX	2.8393 (0.3443)	0.6963 (3.5461)	2.9859 (0.3347)	0.5246 (3.7080)	2.9862 (0.3344)	0.5284 (3.7101)
S&P 500 index	0.0002 (0.0095)	-0.9376 (27.1590)	0.0002 (0.0113)	-0.9152 (23.1786)	0.0002 (0.0113)	-0.9145 (23.1709)

<i>Cryptocurrency specific factors</i>						
Market capitalisation	24.9133	-0.2207	0.0006	-0.9593	24.5041	0.5297
	1.8849	1.6367	0.0503	13.3268	0.5535	2.6769
<i>Global factor</i>						
EPUI US	4.5933	0.1925	4.8465	0.0379	4.8466	0.0372
	(0.6494)	(3.7628)	(0.6260)	(4.1852)	(0.6262)	(4.1839)
Observations	3412		1965		1964	
Sample start date	01.01.14		18.12.17		19.12.17	

Note: Table reports the mean, standard deviation (in parentheses), skewness and kurtosis (in parentheses) values of the variables. Investor attention variables are measured by the natural logarithm of one plus the daily count of Tweets or Bloomberg news for each cryptocurrency. All the other variables except realised volatility are log transformed. The end date of the sample for all cryptocurrencies is 05.05.2023. Data frequency is daily. To achieve stationarity, variables “Addresses with balance greater than $\geq 0.1\%$ of supply, count”, “Addresses with balance greater than $\geq 0.01\%$ of supply, count”, “Mean transaction fee (USD)”, “S&P 500 index” for BTC, “Individual investor attention”, and “S&P 500 index” for XRP are differenced.

Table A3: SVR modelling of cryptocurrency volatility and investor attention

	BTC	ETH	XRP
Mean absolute error (MAE)	0.0098	0.0114	0.0172
Root mean square error (RMSE)	0.0139	0.0168	0.0243
Kernel function	Gaussian	Gaussian	Gaussian
Training performance (RMSE)	0.0142	0.0166	0.0237
Validation performance (RMSE)	0.0136	0.0161	0.0229
Test performance (RMSE)	0.0126	0.0157	0.0227
Kernel function	Gaussian	Gaussian	Gaussian

Note (also for Table A4 below): These models use the same independent (predictor) variables as in Table 3. Several variables are differenced to achieve stationarity. The first three rows report accuracy measures and optimal kernel function for the cross-validation scheme with five cross-validation folds. The second four rows represent accuracy measures and optimal kernel function for the holdout validation scheme with validation and test sizes set as 15% and 15% of the sample, respectively.

Table A4: SVR modelling of cryptocurrency volatility and investor attention including the whale effect and control variables

	BTC		ETH		XRP	
Whale effect ($\geq 0.1\%$ of supply)						
Mean absolute error (MAE)	0.0095	0.0096	0.0098	0.0099	0.0119	0.0121
Root mean square error (RMSE)	0.0127	0.0128	0.0140	0.0141	0.0175	0.0177
Kernel function	Quadratic	Quadratic	Gaussian	Gaussian	Gaussian	Gaussian
Training performance (RMSE)	0.0125	0.0124	0.0129	0.0128	0.0160	0.0161
Validation performance (RMSE)	0.0124	0.0124	0.0131	0.0131	0.0158	0.0158
Test performance (RMSE)	0.0122	0.0122	0.0129	0.0130	0.0162	0.0163
Kernel function	Quadratic	Quadratic	Gaussian	Gaussian	Gaussian	Gaussian
Whale effect ($\geq 0.01\%$ of supply)						
Mean absolute error (MAE)	0.0087	0.0087	0.0099	0.0099	0.0119	0.0122
Root mean square error (RMSE)	0.0119	0.0118	0.0141	0.0142	0.0177	0.0181
Kernel function	Cubic	Cubic	Gaussian	Gaussian	Gaussian	Gaussian
Training performance (RMSE)	0.0109	0.0108	0.0128	0.0125	0.0156	0.0157
Validation performance (RMSE)	0.0110	0.0110	0.0126	0.0121	0.0152	0.0153
Test performance (RMSE)	0.0107	0.0108	0.0129	0.0126	0.0151	0.0154
Kernel function	Cubic	Cubic	Gaussian	Gaussian	Gaussian	Gaussian

Note: These models use the same independent (predictor) variables as in Tables 4 and 5. The models in columns 2, 4, and 6 include an interaction term between the respective whale effect and institutional investor attention variables.

Table A5: RF modelling of cryptocurrency volatility and investor attention

	BTC	ETH	XRP
Mean absolute error (MAE)	0.0098	0.0115	0.0175
Root mean square error (RMSE)	0.0132	0.0162	0.0230
Training performance (RMSE)	0.0122	0.0144	0.0213
Validation performance (RMSE)	0.0125	0.0150	0.0216
Test performance (RMSE)	0.0123	0.0146	0.0211

Note (also for Table A6 below): The table presents results for the bagged decision trees machine learning method, which uses Breiman's (2001) random forest algorithm. These models use the same independent (predictor) variables as in Table 3. The response variable is realised volatility. Several variables are differenced to achieve stationarity. The first two rows report accuracy measures for the cross-validation scheme with five cross-validation folds. The second three rows represent accuracy measures for the holdout validation scheme with validation and test sizes set as 15% and 15% of the sample, respectively. The minimum leaf size is set as 14, and the number of learners is kept at a rate of 30. In MATLAB's bagged decision trees, the default maximum number of decision splits is $n - 1$ where n is the number of observations, while the number of predictors randomly selected for each split is one-third of the total predictors (MathWorks, 2024).

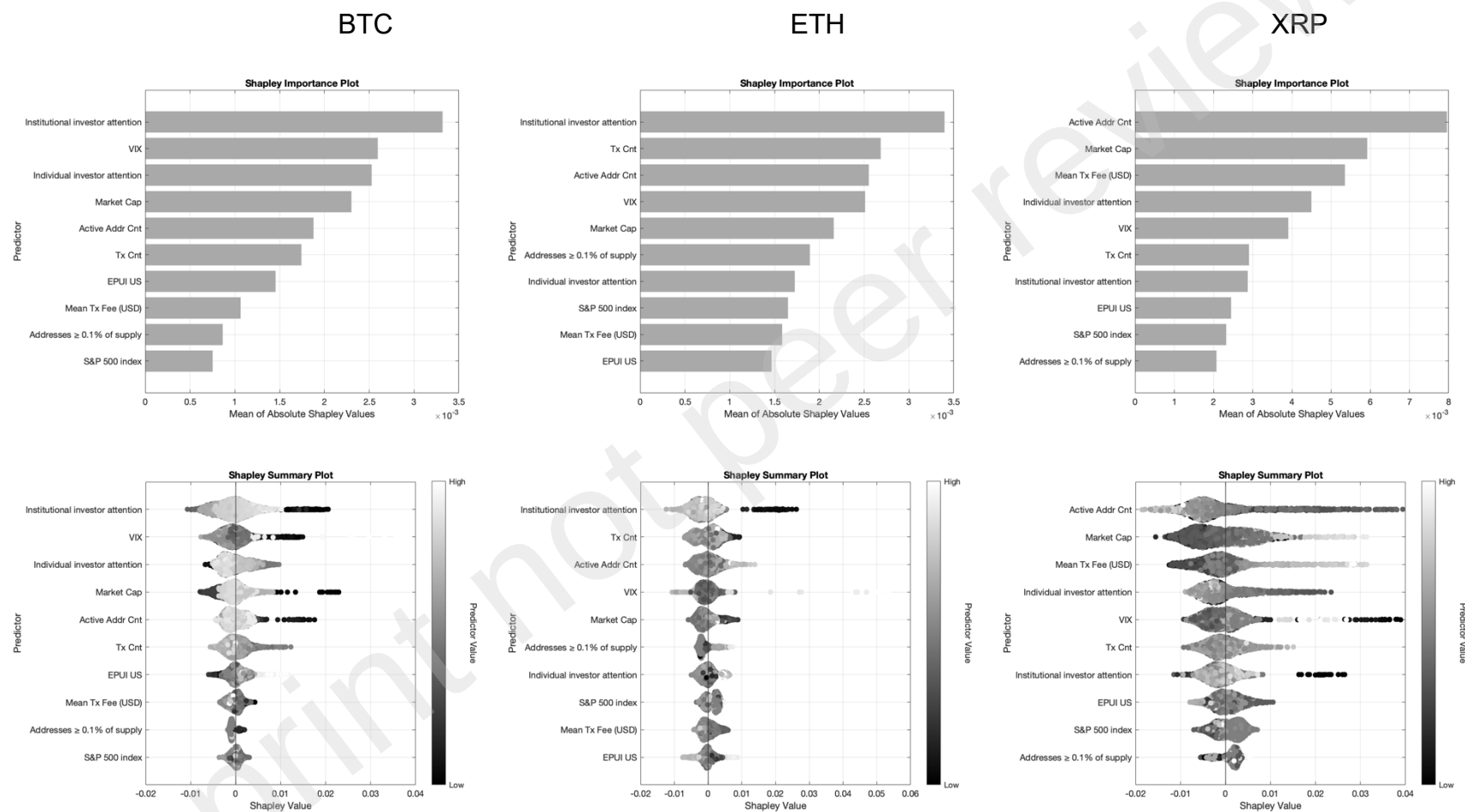
Table A6: RF modelling of cryptocurrency volatility and investor attention including the whale effect and control variables

	BTC		ETH		XRP	
Whale effect ($\geq 0.1\%$ of supply)						
Mean absolute error (MAE)	0.0074	0.0075	0.0090	0.0091	0.0101	0.0103
Root mean square error (RMSE)	0.0098	0.0100	0.0119	0.0120	0.0137	0.0139
Training performance (RMSE)	0.0084	0.0085	0.0099	0.0100	0.0118	0.0121
Validation performance (RMSE)	0.0098	0.0096	0.0116	0.0112	0.0121	0.0123
Test performance (RMSE)	0.0095	0.0095	0.0111	0.0111	0.0126	0.0128
Whale effect ($\geq 0.01\%$ of supply)						
Mean absolute error (MAE)	0.0066	0.0067	0.0090	0.0091	0.0102	0.0105
Root mean square error (RMSE)	0.0089	0.0091	0.0119	0.0120	0.0138	0.0141
Training performance (RMSE)	0.0076	0.0075	0.0099	0.0101	0.0115	0.0123
Validation performance (RMSE)	0.0089	0.0089	0.0117	0.0114	0.0123	0.0128
Test performance (RMSE)	0.0088	0.0085	0.0108	0.0113	0.0125	0.0127

Note: These models use the same independent (predictor) variables as in Tables 4 and 5. The models in columns 2, 4, and 6 include an interaction term between the respective whale effect and institutional investor attention variables.

Figure A1: Shapley importance and summary plots from RF with investor attention, whale effect and control variables

a) Whale effect ($\geq 0.1\%$ of supply)



b) Whale effect ($\geq 0.01\%$ of supply)

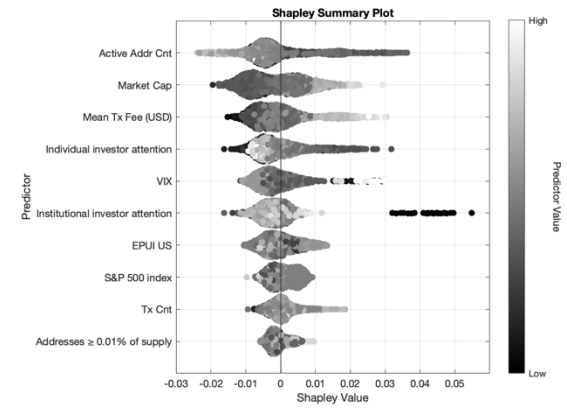
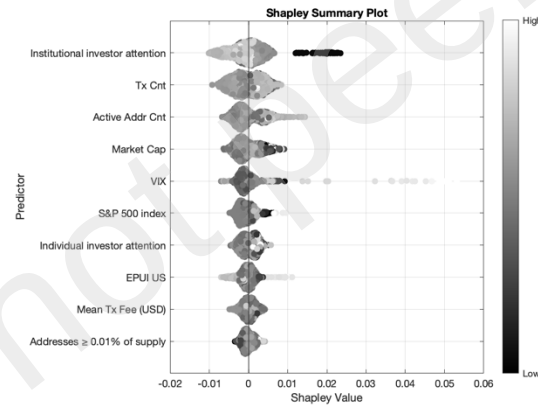
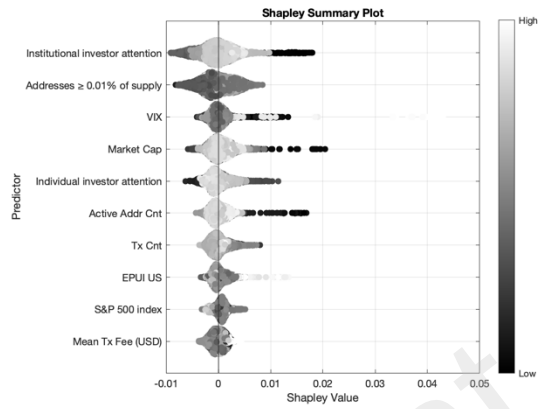
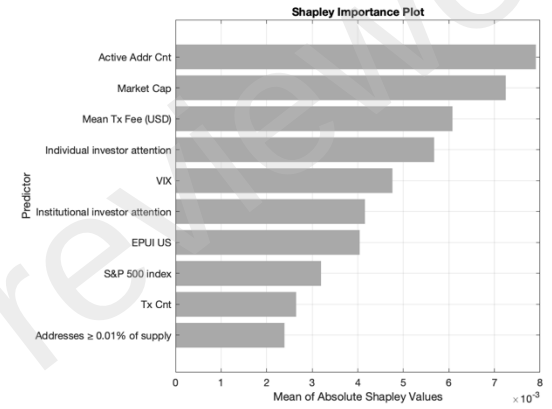
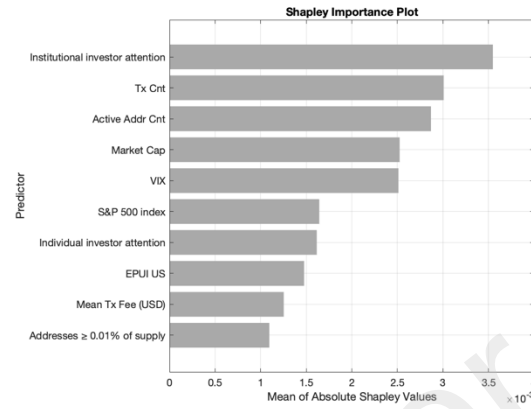
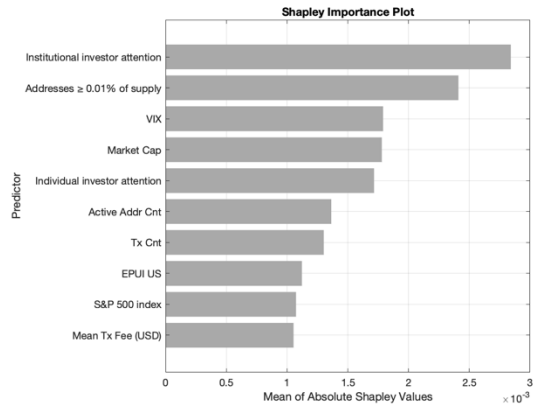
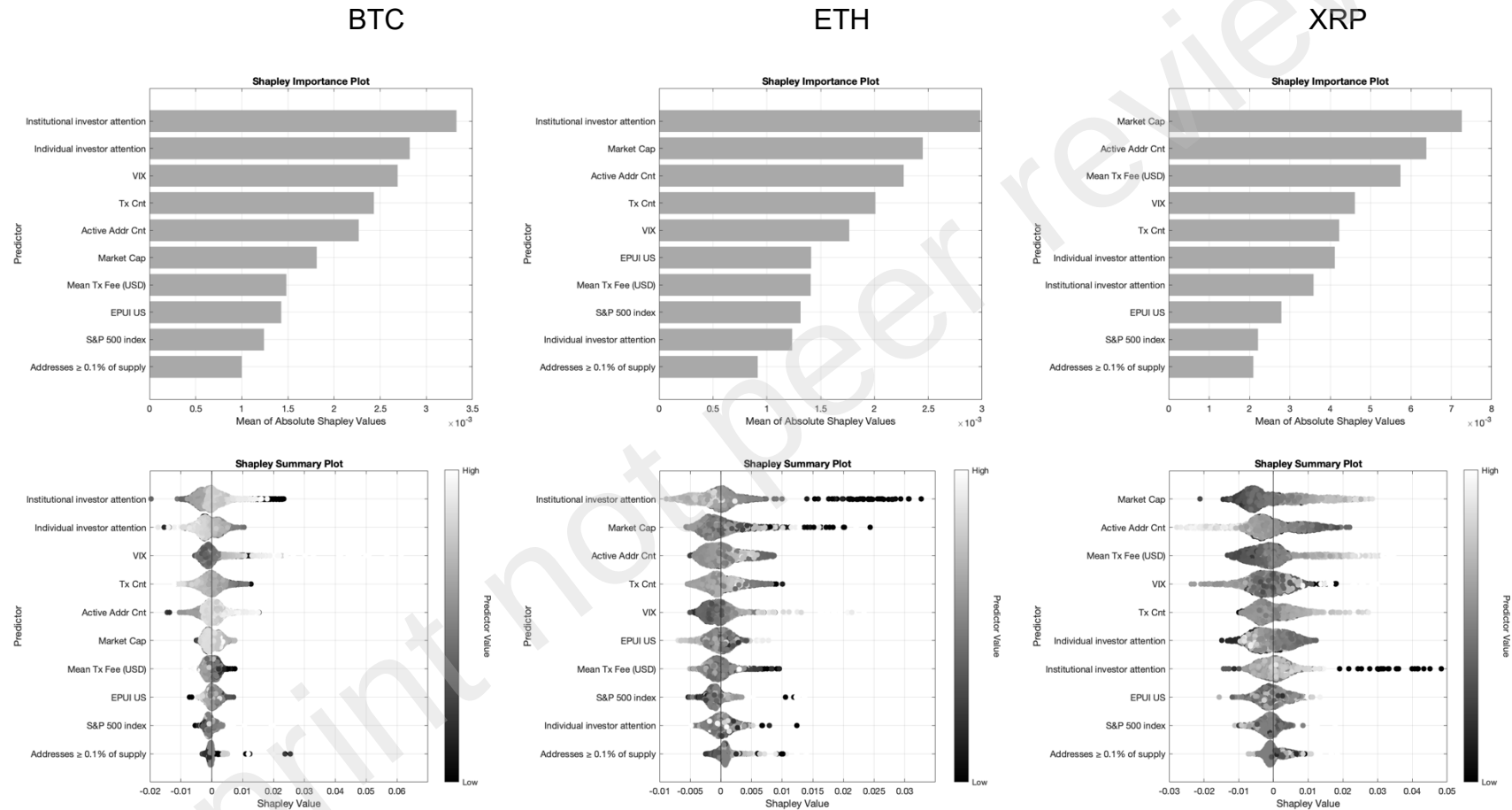


Figure A2: Shapley importance and summary plots from SVR with investor attention, whale effect and control variables

a) Whale effect ($\geq 0.1\%$ of supply)



b) Whale effect ($\geq 0.01\%$ of supply)

